Analyse IV
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Contents

Introduction

Overview of the course The course is a 4cts course taken by computer science, communication system, chemistry, environmental science. The course covers the following topics:

- Complex analysis, complex differentiation, complex integration, Laurent series, residue theorem
- Review of Fourier transform/series
- Laplace transform
- Applications to ODES/PDES (initial and Cauchy value problems)

Main textbook The main book that this course refers to is '*Analyse avancée pour ingénieur*' by Dacorrogna Bernard and Tanteri Chiara.

Part I

Complex Numbers

Chapter 1

Review of Complex number $\mathbb{C}$

The goal of this chapter is to review what we have seen in Analysis I in order to be more comfortable when introducing new materials. As an addition I will add a bit of context on where and why complex analysis differs from multivariate analysis (this is not in the course but is always nice to have in order to better understand the materials).

Imaginary unit As seen in previous classes, we define i the **imaginary unit** such that:

$$i^2 = -1$$

We also define $z \in \mathbb{C}$ as:

$$z = x + iy \qquad x, y \in \mathbb{R}$$

Furthermore:

$$\begin{aligned} x &= \mathrm{Re} z \quad \text{real part} \\ y &= \mathrm{Im} z \quad \text{imaginary part} \end{aligned}$$

The field $\mathbb{C}$ can by fully mapped to $\mathbb{R}^2$, what this means is that every element in $\mathbb{C}$ can be directly '*translated*' to $\mathbb{R}^2$ **and** vis versa.

Complex Conjugate

Definition 1. Let us define the complex conjugate of z as:

$$\overline{z} = x - iy$$

Absolute Value

Definition 2. The absolute value (modulus) is defined as:

$$|z| = \sqrt{x^2 + y^2}$$

> *Intuition* The reason for this definition really comes from Pythagoras theorem and the L_2 norm in $\mathbb{R}^2$.

Furthermore $|z_0 - z_1| = d(z_0, z_1)$ (the distance between those two number).
We can define domain in $\mathbb{C}$ as:

$$Q = \{z \in \mathbb{C} : |z - (1 + i)| < 1\}$$

Which is the disk centred in $z = 1 + i$ of radius 1.

Euler's Formula

Definition 3. For $\theta \in \mathbb{R}$:

$$e^{i\theta} = \cos\theta + i \sin\theta$$

So $e^{i\theta}$ lies on the unit circle around $z = 0$ (because $|e^{i\theta}| = 1$)

Definition 4. The Complex conjugate of the Euler formula is defined as

$$\begin{aligned}\overline{e^{i\theta}} &= \overline{\cos\theta + i \in \theta} \\ &= \cos\theta - i\sin\theta \\ &= \cos(-\theta) + i\sin(-\theta) \\ &= e^{-i\theta}\end{aligned}$$

Geometric intuition What happens when doing the complex conjugate (if we see $\mathbb{C}$ as a plan in $\mathbb{R}^2$) is a symmetry around the x axis.

And this totally makes send when we recall on how it was defined when $z = x + iy$, $\overline{z} = x - iy$ we map the Im part to his opposite.

Remark 1. The rigorous definition of e^z is by the Taylor series. One should more see it as exp being **defined** by the Taylor serie and not the other way around. There is a great video about the definition of the exponential by 3Blue1Brown about taking the exponent of a matrix.

Algebraic Operations Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ we have the followings:

$$\begin{aligned}z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2)\end{aligned}$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

$$\begin{aligned}z_1 \cdot z_2 &= (x_1 + iy_1) \cdot (x_2 + iy_2) \\ &= x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2)\end{aligned}$$

$$z \cdot \overline{z} = x^2 + y^2 = |z|^2 \in \mathbb{R}$$

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{\overline{z_2}z_1}{\overline{z_2}z_2} \\ &= \frac{\overline{z_2}z_1}{|z_2|^2} \qquad \text{if } z_2 \neq 0\end{aligned}$$

Polar coordinates

- If $z \in \mathbb{C}$ then $z = |z|e^{i\theta}$ for some $\theta \in \mathbb{R}$.
- This also means that $z^k = |z|^k e^{ik\theta}$ for $k \in \mathbb{Z}$

Let us define now our two complex numbers in polar coordinates $z_1 = |z_1|e^{i\theta_1}, z_2 = |z_2|e^{i\theta_2}$. If we want to apply algebraic operations defined previously to our new notation we would have:

$$z_1 \cdot z_2 = |z_1||z_2|e^{i(\theta_1+\theta_2)}$$

Dividing our two numbers really comes down to multiply by the inverse which is defined as $z_2^{-1} = \frac{1}{|z_2|}e^{-i\theta_2}$

Lastly:

$$z_1 = z_2 \iff |z_1| = |z_2| \wedge \theta_1 - \theta_2 = 2\pi \cdot k, \quad k \in \mathbb{Z}$$

Geometric Intuition Multiplication by $e^{i\theta_0}$ corresponds to a rotation by an angle θ_0.

Remark 2. Therefore as we can see any number in $\mathbb{C}$ requires only two variables, his norms and argument, which means that when wanting to express any complex number, one only need to have the angle (the rotation) and the modulus. This can be very interesting when our goal is to visualize complex functions. In order to have a good visualization, we need 4 dimensions: two for the input, two for the output. But if all we need is the angle $\theta \in [-\pi, \pi]$ and $|z| \in \mathbb{R}_+$ then we can map using colors for the angle and the third axis for the modulus. This way we can directly see where our number collaps to 0 or when it goes to $\pm\infty$. (there is a great youtube video about 5 ways to visualize complex functions that I recommend).

The 5 ways to visualize complex functions | Essence of complex analysis 3

Principal argument

Definition 5. Let $z = |z|e^{i\theta} \in \mathbb{C}$, the principal argument is $]\theta' \in -\pi, \pi]$ such that:

$$z = |z|e^{i\theta'}$$

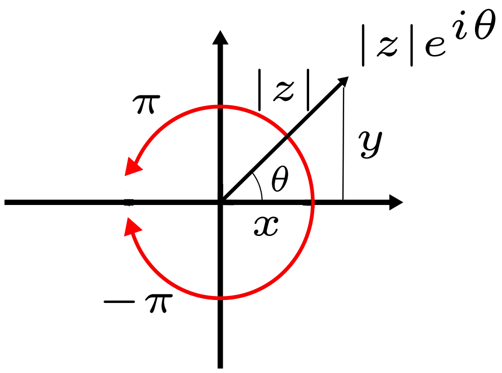


Figure 1.1: Polar representation

Example Let us try to solve the following equation:

$$z^5 = 2$$

To do so let us write our number z into polar coordinate: $z = |z|e^{i\theta}$. If we insert our number into our equation we have:

$$2 = |z|^5 e^{i5\theta}$$

As $|e^{i5\theta}| = 1$ we have that

$$2 = |z|^5 \implies |z| = 2^{\frac{1}{5}}$$

And as 2 is a real number this implies that $5\theta = 2\pi k \implies \theta = \frac{2\pi k}{5}$ where $k \in \mathbb{Z}$ and if we want to take the principal arguments, we have as solution:

$$\theta \in \{-\frac{4\pi}{5}, -\frac{2\pi}{5}, 0, \frac{2\pi}{5}, \frac{4\pi}{5}\}$$

More generally, a k-th order equation will gives us k solutions.

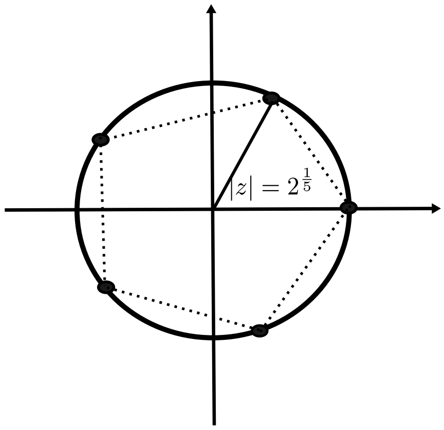


Figure 1.2: Solution of $z^5 = 2$

1.1 Construction of $\mathbb{C}$ (not in the course)

The goal of this section is to add a bit more context on the $\mathbb{C}$ field. First let us recall how real analysis worked. We used the field $(\mathbb{R}, +, \cdot)$ which is an **ordered** field, i.e. we have an ordering on our elements. Formally:

Definition 6. Ordered field

$\forall x, y \in \mathbb{R}$ precisely one of $x < y, x > y, x = y$ holds true.

- $x, y > 0 \implies x \cdot y > 0$

This is the axiomatic definition of an ordered field.

And as we already know, $\mathbb{C}$ is not an ordered field, $2 + i$ cannot be greater nor smaller to 12.

One of the reason on why $\mathbb{C}$ has come up is that the field $(\mathbb{R}, +, \cdot)$ is **not algebraically closed**. We can find polynomials with real coefficient which don't have real roots. $x^2 + 1$ has not root over $\mathbb{R}$. This is a major issue, we need to find a field which is algebraically closed. The question now that was asked: does $\left(\mathbb{R}^2, +\right)$ (which is already an abelian group) can be turned into a field $\left(\mathbb{R}^2, +, ?\right)$? And what about $(\mathbb{R}^n, +)$ is there a way to 'create' a multiplication which would be suitable?

For $n = 1$ this is just the real number. For $n = 2$ the resulting field you will get is $\mathbb{C}$. For $n = 4$ the answer is the Hamilton group (kind of) and the same for $n = 8$. For all other number it is just **impossible**! And this is pretty bizarre, it worked for $1, 2, 4, 8$ and then it stopped.

Theorem 1. If we can create a field on $\mathbb{R}^2$ then

$$\exists z \in \left(\mathbb{R}^2, +, \cdot\right) \text{ s.t. } z^2 = -1$$

Proof. *Let us choose a basis element* $\{1, e\}$ *of* $\mathbb{R}^2$. *This directly implies that:*

$$z = x \cdot 1 + y \cdot e \text{ , for some } x, y \in \mathbb{R}$$

If we want to multiply our number by itself:

$$z^2 = x^2 \cdot 1 + 2xy \cdot e + y^2 \cdot e^2$$

Remark 3. A strange thing we did here is to add the $2xy \cdot e$. Recall that previously, when we had a vector in $\mathbb{R}^2, x$ that we squared, the output that we will get would be:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + x_2^2$$

Or if we have written them using the basis vector $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ this would gives us:

$$(x_1e_1 + x_2e_2) \cdot (x_1e_1 + x_2e_2)$$

The important thing is that both equation above are **totally equivalent**. If we develop this we will get:

$$x_1^2e_1^2 + 2x_1x_2e_1e_2 + x_2^2e_2^2$$

And because e_1 and e_2 are orthogonal, by definition $e_1e_2 = 0$:

$$= x_1^2e_1^2 + x_2^2e_2^2 = x_1^2 + x_2^2$$

Now we want to rewrite e^2 using the basis that we introduced before:

$$e^2 = a \cdot 1 + b \cdot e, \qquad a, b \in \mathbb{R}$$

This implies that:

$$z^2 = \left(x^2 + ay^2\right) \cdot 1 + \left(2xy + by^2\right) \cdot e$$

Now we want to show that there exists the pair $(x, y) \in \mathbb{R}^2$ such that $z^2 = -1$. Then we must have the following:

$$2xy + by^2 = 0 \iff \begin{cases} y = 0 \\ x = -\frac{b}{2}y \end{cases} \quad \textit{or}$$

But here $y = 0$ is not sufficient because this implies for the second relation:

$$x^2 + ay^2 = -1$$

that $x^2 = -1$ where $x \in \mathbb{R}$ which is impossible. We therefore look into the second case, $y \neq 0$ and $x = -\frac{b}{2}y$. In this case this implies that:

$$\begin{aligned} z^2 &= \left(-\frac{b}{2}y\right)^2 + ay^2 \cdot 1 + \underbrace{\left(2\left(\frac{-b}{2}\right)y^2 + by^2\right) \cdot e}_{=0} \\ &= \left(a + \frac{b^2}{4}\right)y^2 \cdot 1 \end{aligned}$$

The second claim is that $a + \frac{b^2}{4} < 0$

Proof. *The proof will be done by contradiction. Suppose that $a + \frac{b^2}{4} \geq 0$. This implies that we can take the square root (in the sense of $\mathbb{R}$) and therefore we have that:*

$$z = \pm y\sqrt{a + \frac{b^2}{4}}$$

Which implies that:

$$\left(z - y\sqrt{a + \frac{b^2}{4}}\right)\left(z + y\sqrt{a + \frac{b^2}{4}}\right) = 0$$

This implies that either $z = y\sqrt{a + \frac{b^2}{4}} = x \cdot 1 + y \cdot e$ *or the same with a negative sign.*

$$\pm y\sqrt{a + \frac{b^2}{4}} \cdot 1 = x \cdot 1 + y \cdot e$$

Since $\{1, e\}$ *are linearly independent, this equality is impossible unless* $y = 0$ *which contradicts the case we are studying* ($y \neq 0$). *Therefore* $a + \frac{b^2}{4} < 0$ □

Now our goal is to proof there exists a y *such that* $\left(a + \frac{b^2}{4}\right)y^2 = -1$. *We can set*

$$y := \frac{1}{-\sqrt{a + \frac{b^2}{4}}}, \qquad x = \frac{-b}{2}y$$

This directly implies that:

$$\begin{aligned} z^2 &= \left(a + \frac{b^2}{4}y^2\right) \\ &= \left(a + \frac{b^2}{4}\right)\left(\frac{-1}{a + \frac{b^2}{4}}\right) = -1 \end{aligned}$$

□

The claim we just prove is not trivial at all, if we want to turn $\mathbb{R}^2$ into a field, then there has to be an element of the field such that $d = -1$. This also implies that our multiplication will be a new one which will be pretty counter-intuitive for us. Now let us pick our element $z \in \left(\mathbb{R}^2, +, \cdot\right)$ such that $z^2 = -1$, and let us call it **i**.

Now we can define our field ℂ:

Definition 7. Let us define the **complex field** ℂ:

$$\begin{aligned} \mathbb{C} &:= \mathbb{R}[i] := \{a + bi, \quad a, b \in \mathbb{R}\} \\ &= \mathbb{R}[X]/(X^2 + 1) := \{a + bX \in \mathbb{R}[X]\} \end{aligned}$$

ℂ is algebraically closed

Theorem 2. Any $P \in \mathbb{C}[X]$ non-zero (not the zero polynomial) has at least a root in ℂ.

Proof. *To prove this we will do it by contradiction. Assume that* P *is a non-constant polynomial and suppose it* ***has no root*** *in* ℂ. *Then this implies that:*

$$P(z) \neq 0 \quad \forall z \in \mathbb{C}$$

Therefore, we can define:

$$f(z) = \frac{1}{P(z)}$$

Since $P(z) \neq 0$ *everywhere and* P *is holomorphic,* f *is also holomorphic on all of* ℂ *(entire).*

The next step of our proof is to show that f *is bounded. Let*

$$P(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_0$$

For large $\|z\|$, *the leading term dominates:*

$$\|P(z)\| \approx \|a_n\|\|z\|^n$$

More precisely, there exists R such that if $\|z\| > R$ then:

$$\|P(z)\| \geq \frac{\|a_n\|}{2}\|z\|^n$$

Thus:

$$\|f(z)\| = \frac{1}{\|P(z)\|} \leq \frac{2}{\|a_n\|\|z\|^n}$$

Therefore, when $\|z\| \to \infty$ we have that

$$f(z) \to 0$$

Hence f is bounded outside a large disk. And inside $\|z\| < R$, the continuous function f on a compact set is also ***bounded****. Therefore, f is bounded on all of $\mathbb{C}$.*

All the work is done now, all we have to do is to apply Liouville's theorem:

Theorem 3. Every bounded entire function is constant

Which therefore implies that f must be constant. But we just said

$$f(z) = \frac{1}{P(z)}$$

which means that P is constant which is a contradiction.

Conclusion:

Every non-constant polynomial in $\mathbb{C}[x]$ has a complex root.

$\square$

Part II

Holomorphic Functions

Chapter 2

Complex Functions

Definition 8. We call f a complex function as:

$$\begin{aligned} f : \mathbb{C} &\to \mathbb{C} \\ z &\mapsto f(z) = f(x+iy) \\ &\qquad\quad = u(x,y) + iv(x,y) \end{aligned}$$

In particular we have $u, v : \mathbb{R}^2 \to \mathbb{R}$

A nice thing to see here is the relation between $\mathbb{R}^2$ and $\mathbb{C}$. As said before we can always map a complex number to a number in $\mathbb{R}^2$. By 'splitting' our function into to two sub functions u, v which map to $\mathbb{R}$. We effectively transformed our function $f : \mathbb{C} \to \mathbb{C}$ into two sub function that goes from $\mathbb{R}^2 \to \mathbb{R}$ which therefore point to $\mathbb{R}^2$ (if we replaced the 1 by a unit vector, and i as another unit vector).

Examples First let us see the simplest example, the **unit** function: $f(z) = z = x + iy$. In this case we can easily see that

$$u(x,y) = x \qquad v(x,y) = y$$

A bit more sophisticated function could be for instance: $f(z) = \overline{z} = x - iy$. Which gives us:

$$u(x,y) = x \qquad v(x,y) = -y$$

If we take the square we have $f(z) = z^2 = (x+iy)^2 = \left(x^2 - y^2\right) + i2xy$

In all of those expression we can easily see where our function u and v are. But is it always the case?

Trigonometric Functions

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Remark 4. We have already seen the same concept in Analysis III. We use a lot those definition when dealing with Fourier series. (with odd/even series)

$$\cosh z = \frac{e^z + e^{-z}}{2}$$

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

The rigorous definition of e^z is by the Taylor series. One should more see it as exp being **defined** by the Taylor serie and not the other way around. There is a great video about the definition of the exponential by 3Blue1Brown.

2.1 Limit and Continuity

Definition 9. We say $f : \mathbb{C} \to \mathbb{C}$ is **continuous** at a point $z_0 \in \mathbb{C}$ if $\forall \varepsilon > 0$ there exists $\delta > 0$ such that $\forall z \in \mathbb{C}$ with $\|z - z_0\| \leq \delta$ we have:

$$\|f(z) - f(z_0)\| \leq \varepsilon$$

Equivalently:

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Remark 5. This definition is directly mapped to the one seen in Analysis II. What we are really doing here in fact is the limit of our function the same way as we would have done in $\mathbb{R}^2$. If we develop this:
Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ then f is continuous at $z_0 = x_0 + y_0 i$ if and only if u and v are continuous at (x_0, y_0)

Operation on Continuous Functions

If f, g are continuous functions then the same is true for:

- $f + g$
- $f \cdot g$
- $\overline{f}$
- $f \circ g$
- $\vdots$

2.2 Complex Derivative

Definition 10. We say $f : \mathbb{C} \to \mathbb{C}$ is **complex differentiable** at a point $z_0 \in \mathbb{C}$ if:

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \text{ exists}$$

We denote this limit by $f'(z_0)$

Definition 11. If f is differentiable at every point of an open set $\omega \subseteq \mathbb{C}$, we say f is **holomorphic** on ω.

Remark 6. In many books about complex analysis we use the term analytic instead of holomorphic, both terms are equivalent for function of complex variable, but this equivalence is a deep theorem and not a definition. The definition 11 here is the definition of **holomorphic**. On the other hand, analytic on ω means that around every point $z_0 \in \omega$, f can be represented by a convergent power series:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

In some neighbourhood of z_0. This is an *a priori* much stronger condition.

The fact that f is holomorphic $\iff$ f is analytic (on ω) can be proven, and we will be able to do prove it.

The Analytic $\implies$ holomorphic proof is pretty straightforward: as f can be represented by a power series and that a power series is differentiable (within their radius of convergence) then f is also differentiable the same way.

For the holomorphic $\implies$ Analytic proof. This is harder and we will go through it later when we will have introduced Cauchy's integral formula

This theorem doesn't work in $\mathbb{R}$ a function can be infinitely differentiable (smooth) on $\mathbb{R}$ without being analytic. For instance:

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

This is C^∞ everywhere, but its Taylor series at 0 is identically 0, so it fails to represent f near 0

Definition 12. If f is holomorphic on $\mathbb{C}$, we call f **entire**.

Remark 7. The same rules of differentiation of products (product rule, Leibniz rule), Compositions (chain rule), etc. are as in the real case.

Recall on topology
Open set

Definition 13. $\omega \subseteq \mathbb{C}$ is **open** if for all $z_0 \in \omega$, there exists $\Gamma > 0$ such that the disk $D_\Gamma(z_0) = \{z \in \mathbb{C} : \|z - z_0\| < r\} \subseteq \omega$

Examples

- $\mathbb{C}$ is open
- $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is open
- $\{z \in \mathbb{C} : \text{Re } z \geq 0\}$ is not open

Examples of Differentiation

Let $f(z) = z$ our goal is to compute the derivative of f for all $z_0 \in \mathbb{C}$. By the definition:

$$f'(z_0) = \lim_{z \to z_0} \frac{z - z_0}{z - z_0} = 1$$

In particular, f is (holomorphic and) entire.
Let us try a little more sophisticated example. Let $f(z) = \frac{1}{z}$. For all $z_0 \in \mathbb{C}^*$:

$$f'(z_0) = -\frac{1}{z_0^2}$$

So f is holomorphic on $\mathbb{C}^*$ but not entire.

2.3 Cauchy-Riemann equations

The goal of those equations is to check if a function f is differentiable or/and is holomorphic. When we have quite 'easy' function (as seen before) showing that the function is holomorphic by taking the limit is possible. However when function become tedious we will use Cauchy-Riemann questions:

Definition 14. Let $w \subseteq \mathbb{C}$ open, $f : w \to \mathbb{C}$ let:

$$z = x + iy \mapsto f(z) = u(x, y) + v(x, y)i$$

Be holomorphic. Then, $u, v \in C^1(w)$ (the space of differentiable function $w \subseteq \mathbb{R}^2 \to \mathbb{R}$) with continuous derivative) and:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Those equation are called Cauchy-Riemann equations.

Theorem 4. If the relation:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

holds then f is holomorphic

Proof of the direct implication '$\implies$' Suppose that f is holomorphic on w, therefore, $\forall z_0 \in w$ we have:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{exists}$$

We will calculate the limit in two ways.

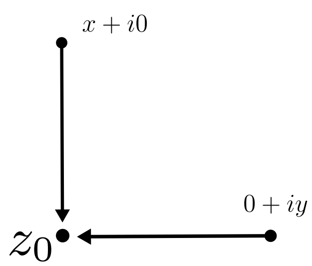


Figure 2.1: Two ways of approaching

First we will approaching z_0 from the x-direction:

$$\begin{aligned} f'(z_0) &= \lim_{x\to x_0} \frac{f(x+y_0i)-f(x_0+y_0i)}{x+y_0i-(x_0+y_0i)} \\ &= \lim_{x\to x_0} \frac{u(x,y_0)+iv(x,y_o)-u(x_0,y_0)-iv(x_0,y_0)}{x-x_0} \\ &= \underbrace{\lim_{x\to x_0} \frac{u(x,y_0)-u(x_0,y_0)}{x-x_0}}_{=\frac{\partial u}{\partial x}(x_0,y_0)} + \underbrace{\lim_{x\to x_0} \frac{v(x,y_0)-v(x_0,y_0)}{x-x_0}i}_{=\frac{\partial v}{\partial x}(x_0,y_0)} \end{aligned}$$

Now we will approach z_0 from the $y-$direction:

$$\begin{aligned} f'(z_0) &= \lim_{y\to y_0} \frac{f(x_0,yi)-f(x_0+y_0i)}{x_0+yi-(x_0+y_0i)} \\ &= \lim_{y\to y_0} \frac{u(x_0,y)+iv(x_0,y)-u(x_0,y_0)-iv(x_0,y_0)}{i(y-y_0)} \\ &= \overbrace{\frac{1}{i}}^{=-i} \underbrace{\lim_{y\to y_0} \frac{u(x_0,y)-u(x_0,y_0)}{y-y_0}}_{=\frac{\partial u}{\partial y}(x_0,y_0)} + \frac{i}{i} \underbrace{\lim_{y\to y_0} \frac{v(x_0,y)-v(x_0,y_0)}{y-y_0}}_{=\frac{\partial v}{\partial y}(x_0,y_0)} \end{aligned}$$

By comparing real and imaginary part leads to the Cauchy-Riemann equations. Moreover form this proof, we get explicit expression for $f'(z_0)$:

$$f' = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}i = \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}i$$

Example $f : \mathbb{C}^* \to \mathbb{C}$, $f(z) = \frac{1}{z}$

$$\begin{aligned} f(z) &= \frac{\overline{z}}{\|z\|^2} \\ &= \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i \\ &= u(x,y) + v(x,y)i \end{aligned}$$

Now we will compute the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} \qquad \frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2} \qquad \frac{\partial v}{\partial y} = \frac{-\left(x^2+y^2\right)+2y^2}{(x^2+y^2)^2}$$

We therefore the Cauchy-Riemann equations satisfied and:

$$\begin{aligned} f' &= \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}i \\ &= \frac{y^2-x^2}{(x^2+y^2)^2} + \frac{2xy}{(x^2+y^2)^2}i \end{aligned}$$

Indeed, this coincides with $-\frac{1}{z^2}$ (computation).

Example The last example will be: $f(z) = e^z$ let us massage our function:

$$\begin{aligned} f(z) &= e^{x+yi} = e^x e^{yi} \\ &= e^x \cos y + e^x \sin(y)i \\ &= u(x,y) + iv(x,y) \end{aligned}$$

Not let us compute the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = e^x \cos y \qquad \frac{\partial v}{\partial x} = e^x \sin y$$
$$\frac{\partial u}{\partial y} = -e^x \sin y \qquad \frac{\partial v}{\partial y} = e^x \cos y$$

As we can see that equation holds which implies that our function f is holomorphic.

Moreover, the derivative of our function is given as:

$$f' = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^z$$

2.4 Rules for differentiation

Theorem 5. Let $f, g : w \to \mathbb{C}$ holomorphic. Then:

$$\begin{aligned} (f+g)' &= f' + g' && \\ (fg)' &= f'g + fg' && \text{Leibniz rule} \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} && g \neq 0 \end{aligned}$$

All of these relations can be shown exactly as in the real case, we will do $(fg)'(z_0)$. First let us use the definition of the derivative:

$$\begin{aligned} (fg)'(z_0) &= \lim_{z\to z_0} \frac{f(z)g(z) - f(z_0)g(z_0)}{z - z_0} \\ &= \lim_{z\to z_0} \frac{f(z)g(z) - f(z_0)g(z) + f(z_0)g(z) - f(z_0)g(z_0)}{z - z_0} \\ &= \lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0} g(z) + f(z_0)\frac{g(z) - g(z_0)}{z - z_0} \\ &= f'(z_0)g(z_0) + f(z_0)g'(z_0) \end{aligned}$$

With these rules and previous examples, it follows the usual formulas for differentiation of polynomial, trigonometric functions, etc...

$$\begin{aligned} \left(z^k\right)' &= kz^{k-1} \\ (\cos z)' &= -\sin z \\ (\sin z)' &= \cos z \end{aligned}$$

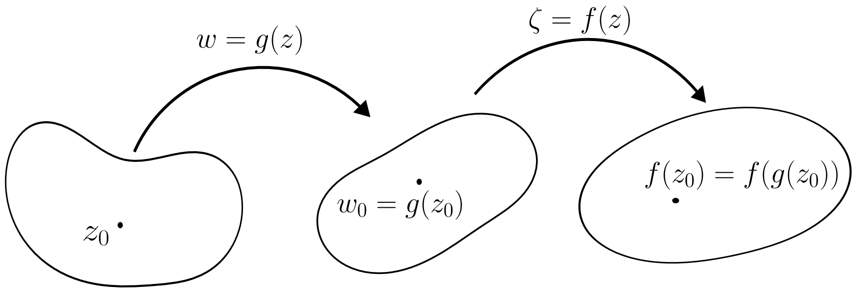
The chain rule is also valid for the complex derivative. We give careful statement and proof

Theorem 6. Suppose that $g(z)$ is differentiable at z_0, and suppose that $f(w)$ is differentiable at $w_0 = g(z_0)$. Then the composition $(f \circ g)(z) = f(g(z))$ is differentiable at z_0 and:

$$(f \circ g)'(z_0) = f'(g(z_0)) \cdot g'(z_0)$$

Remark A useful mnemonic device for remembering the chain rule is:

$$\frac{df}{dz} = \frac{df}{dw}\frac{dw}{dz}$$



Geometric interpretation of Cauchy-Riemann equations

Let's fix $f : \mathbb{C} \to \mathbb{C}$ holomorphic, and view it as a function $f : \mathbb{R}^2 \to \mathbb{R}^2$. Geometrically the Jacobian matrix:

$$Df = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{pmatrix}$$

tells us how f acts on 'infinitesimal rectangles'/ disturb volume.

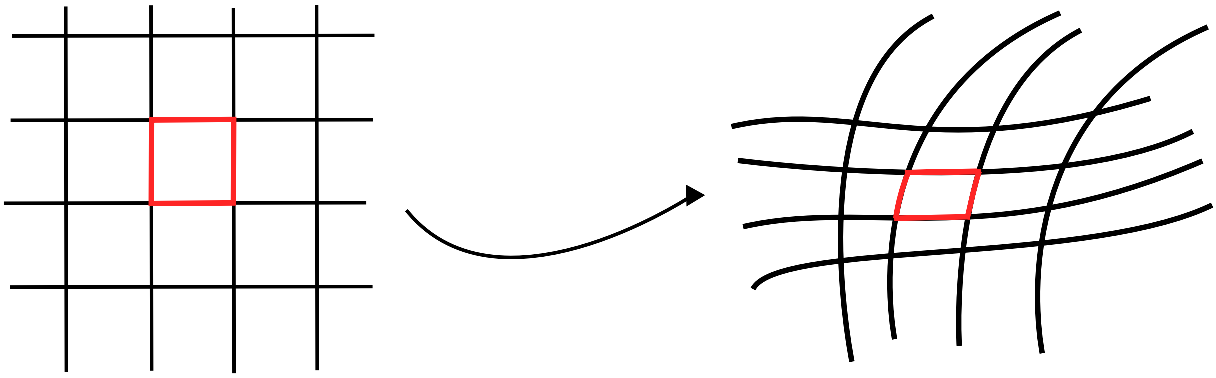


Figure 2.2: Transformation of the Jacobian matrix

To go back on what I have talked about in the Remark 2, this is exactly one of the five way in order to "*visualize*" how function of complex variables look like. Here is the part of the video that talks about it:

The 5 ways to visualize complex functions | Essence of complex analysis 3, z, w plane

A very specific deformation happens for holomorphic functions. In fact, the Cauchy-Riemann equations imply that the Jacobian is in the form of:

$$\begin{aligned} Df &= \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \\ &= \sqrt{a^2+b^2} \underbrace{\begin{pmatrix} \frac{a}{\sqrt{a^2+b^2}} & -\frac{b}{\sqrt{a^2+b^2}} \\ \frac{b}{\sqrt{a^2+b^2}} & \frac{a}{\sqrt{a^2+b^2}} \end{pmatrix}}_{\text{rotation matrix}} \end{aligned}$$

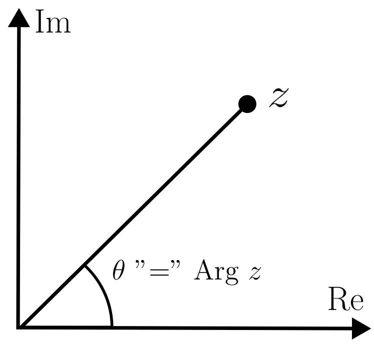
Recall that a rotation matrix is in the form of:

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

Hence we have that $Dg = \lambda\cdot$rotation! Therefore, infinitesimal square is mapped to a square.

2.5 Complex logarithm

First let us recall what the principal argument is. For $z \neq 0$ we have that $-\pi < \text{Arg } z \leq \pi$



It can be computed by:

$$\text{Arg} z = \begin{cases} \arctan \frac{y}{x} & \text{if } x > 0 \\ \frac{\pi}{2} & \text{if } x = 0, y > 0 \\ \pi + \arctan \frac{y}{x} & \text{if } x < 0, y \geq 0 \\ -\frac{\pi}{2} & \text{if } x = 0, y < 0 \\ -\pi + \arctan \frac{y}{x} & \text{if } x < 0, y < 0 \end{cases}$$

Arg is (defined yet) **discontinuous** on the negative real axis, it jumps from π to $-\pi$ across the negative real axis.

Okay this is nice but why are we even talking about this here in the first case, we are trying to define the logarithm for complex number. How de we define complex logarithm? First what does it do? It is the inverse of the exponential meaning that we should have $\log e^z = z$. If we expand z into Cartesian form, $\log(e^{x+iy}) = x + iy$.

If $\log z = x + iy$ then we should have $z = e^{x+iy} = e^x e^{yi}$. By polar coordinate, we can also define z as:

$$z = \underbrace{\|z\|}_{=e^x} \underbrace{e^{i\text{Arg}z}}_{=e^{iy}}$$

Remark 8. We are allowed to do so because $z = e^{x+iy}$. In general we don't have $y =$Arg z. Now we have something that could be put into the log:

Definition 15. We define:

$$lgz = lg\|z\| + i\text{Arg}z$$

Complex logarithm, more precisely the principal branch of complex algorithm

Remark 9.
- $\log z$ is not defined at $z = 0$
- $\log z$ is discontinuous on the negative real axis

More precisely, computing $\log z$ for $z = x + iy$ with $x > 0$:

$$\begin{aligned} \log z &= \log \sqrt{x^2 + y^2} + i \arctan \frac{y}{x} \\ &= u(x, y) + iu(x, y) \end{aligned}$$

It is easy to verify that log is holomorphic on $\mathbb{C} \setminus \{Im = 0, Re \leq 0\}$ and $(\log z)' = \frac{1}{z}$

logarithm rules As nice as it seems to be, we still have the same logarithm rules as in the real case namely

$$\log(z_1 z_2) = \log(z_1) + \log(z_2)$$
$$\log(z^\gamma) = \gamma \log(z) \qquad \gamma \in \mathbb{C}$$

Differentiation The principal logarithm function $w = \log z$ is a continuous inverse for $z = e^w$ for $-\pi < \arg w < \pi$. Since e^w is holomorphic and $(e^w)' \neq 0$. We can conclude that $\log z$ is holomorphic. Let us compute its derivative:

$$z = e^{\log z}$$

Using that fact and using the chain rule, let us derives both side of this equality:

$$1 = e^{\log z} \frac{d}{dz}(\log z) = z \frac{d}{dz}(\log z)$$

Thus

$$\frac{d}{dz} \log z = \frac{1}{z}$$

Remark 10. The trick we just used above can be used to determine a lot of other derivative, for instance

$$\frac{d}{dz}\sqrt{z} = \frac{1}{2\sqrt{z}}$$

Let us define

$$w = \sqrt{z}$$
$$w^2 = z$$

Since $\left(w^2\right)' = 2w$ is not zero for $w \neq 0$, the continuous inverse branch $\sqrt{z}$ is analytic. Differentiating $w^2 = z$, we obtain:

$$2w \frac{dw}{dz} = 1, \qquad \frac{dw}{dz} = \frac{1}{2w}$$

Example If $z_1 = z_2 = -1$ then $\log z_1 = \log z_2 = i\pi$. (since $e^{i\pi} = -1$). Yet we have that $\log(z_1 z_2) = 0 \neq \log(z_1) + \log(z_2)$. Similarly

$$\log\left(z_1^2\right) \neq 2\log(z_1)$$

However these rules remain true up to $2\pi i\mathbb{Z}$:

$$\log(z_1 z_2) - \log z_1 - \log z_2 \in 2\pi i\mathbb{Z}$$
$$\log(z^\gamma) - \gamma \log(z) \in 2\pi i\mathbb{Z}$$

Chapter 3

Complex integration

3.1 Line integrals

Let's first recall from vector analysis: if $\vec{F} : \mathbb{R}^2 \to \mathbb{R}^2$ is a vector field such that

$$F(x_1, x_2) = (F_1(x_1, x_2), F_2(x_1, x_2))$$

And $\gamma : [a, b] \to \mathbb{R}^2$ is a C^1 curve with image Γ then:

$$\underbrace{\int_\Gamma F dl}_{\text{line integral}} := \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt$$

Here the orientation is **important**. We will need to recall some facts:

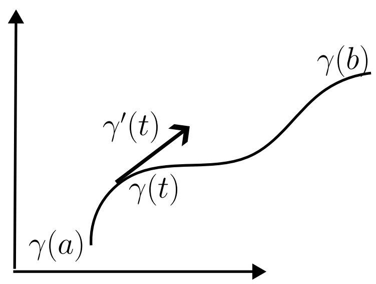


Figure 3.1: Sorry for the curve

1. The line integral is **independent** of the parametrisation, but switching orientation creates a negative sign.
2. If $\gamma : [c, d] \to [a, b]$ is one-to-one, out and increasing, then γ is a parametrization of Γ.
3. If γ is closed, meaning $\gamma(a) = \gamma(b)$ and $\text{curl} F = 0$ $(= \nabla \times F)$ in the 'interior' of Γ, then we have that:

   $$\int_\Gamma F dl = 0$$

   This is a consequence from Stokes theorem
4. A curve is called:
   - **simple** if it has no self intersections
   - **regular** if $\gamma' \neq 0 \; \forall x \in [a, b]$

Now we will define a similar notion of line integrals for complex functions.

Definition 16. Let $\gamma : [a, b] \to \mathbb{C}$ be a regular curve that parametrises $\Gamma \subseteq \mathbb{C}$. Let $f : \Gamma \to \mathbb{C}$ be continuous.

Then we define the line integral of f along Γ by:

$$\int_\Gamma f(z)dz = \int_\gamma f(z)dz := \int_a^b f(\gamma(t)) \cdot \gamma'(t)dt$$

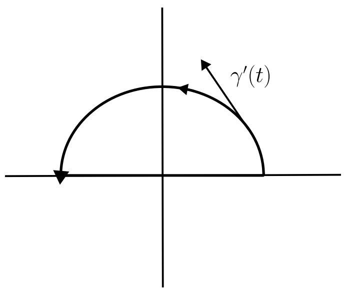
Remark 11. The $\int_\gamma$ is a slight abuse of notation

Usually we write $-\Gamma$ to mean the curve Γ with the **reverse** orientation. In this case we have:

$$\int_{-\Gamma} f(z)dz = -\int_\Gamma f(z)dz$$

Remark 12. The value of this line integral is independent of the parametrization of γ as long as the orientation is preserved.

Example Let Γ be the unit half circle around $z = 0$, parametrized counter clockwise:



$$\begin{aligned} \gamma : [0, \pi] &\to \mathbb{C} \\ \gamma(t) &= e^{it} \\ \gamma'(t) &= ie^{it} \end{aligned}$$

We seek to integrate $f(z) = z^2$ over Γ:

$$\begin{aligned} \int_\Gamma z^2 dz &= \int_0^\pi \left(e^{it}\right)^2 ie^{it}dt \\ &= i\int_0^\pi e^{3it}dt = i\left[\frac{e^{3it}}{ti}\right]_0^\pi = -\frac{2}{3} \end{aligned}$$

Remark Above we used the fundamental theorem of calculus. For more general function, one splits the integrands into real and imaginary part and computes the two real integrals separately.

Example ii Let Γ be the unit circle oriented counter-clockwise. We will use the same γ as before but we will extend its domain of definition:

$$\begin{aligned} \gamma : [0, 2\pi] &\to \mathbb{C} \\ \gamma(t) &:= e^{it} \end{aligned}$$

With again $\gamma'(t) = ie^{it}$. We try to compute $f(z) = z^2$:

$$\begin{aligned}\int_\Gamma z^2 dz &= \int_0^{2\pi} \left(e^{it}\right)^2 ie^{it} dt \\ &= \int_0^{2\pi} e^{3it} dt \\ &= i\left[\frac{e^{3it}}{3i}\right]_0^{2\pi} = 0\end{aligned}$$

This is a special case of the Cauchy integral theorem (that we will see later).

Example iii Let Γ and γ be defined as above. We will take this time $f(z) = \frac{1}{z}$.

$$\begin{aligned}\int_\Gamma \frac{1}{z} dz &= \int_0^{2\pi} \frac{1}{e^{it}} ie^{it} dt \\ &= \int_0^{2\pi} idt = 2\pi i \neq 0\end{aligned}$$

Example iv Now let $f(z) = \frac{1}{z^2}$:

$$\begin{aligned}\int_\Gamma \frac{1}{z^2} &= \int_0^{2\pi} \frac{1}{\left(e^{it}\right)^2} ie^{it} dt = i\int_0^{2\pi} e^{-it} dt \\ &= -\left[e^{it}\right]_0^{2\pi} = 0\end{aligned}$$

3.1.1 Cauchy's Theorem

The Cauchy theorem tells us what the value of a complex integral is on general constraints:

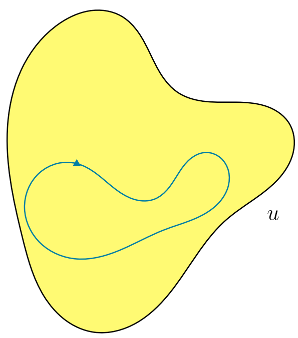
Theorem 7. Let $U \subseteq \mathbb{C}$ open and simply connected set and let Γ be a closed and piecewise regular curve inside U. Let $f : U \to \mathbb{C}$ be holomorphic. Then:

$$\int_\Gamma f(z) dz = 0$$

Geometric intuition

Definition 17. A domain is said simply connected when it is a domain with only one component and '*no holes*' e.g. $\mathbb{C}, \{z \in \mathbb{C} : \|<\| \leq 1\}$, while $\mathbb{C}^*$ is not simply connected.

Definition 18. **Piecewise regular** Γ means Γ might have finitely many curves, e.g. square



Why is this true?

Let $\gamma(t) = \alpha(t) + \beta(t)i$ a parametrization Γ_1 and:

$$f(x + yi) = u(x, y) + iv(x, y)$$

Then the integral gives us:

$$\begin{aligned}
\int_{\Gamma} f(z)dz &= \int_a^b f(\gamma(t)) \cdot \gamma'(t)dt \\
&= \int_a^b [u(\gamma(t)) + iv(\gamma(t))] \cdot [\alpha'(t) + \beta'(t)i]dt \\
&= \int_a^b [u(\gamma(t))\alpha'(t) - v(\gamma(t)) - \beta'(t)]dt \\
&+ i\int_a^b [u(\gamma(t))\beta'(t) + v(\gamma(t))\alpha'(t)]dt \\
&= \int_a^b \begin{pmatrix} u(\gamma(t)) \\ -v(\gamma(t)) \end{pmatrix} \cdot \begin{pmatrix} \alpha'(t) \\ \beta'(t) \end{pmatrix} dt + i\int_a^b \begin{pmatrix} v(\gamma(t)) \\ u(\gamma(t)) \end{pmatrix} \cdot \begin{pmatrix} \alpha'(t) \\ \beta'(t) \end{pmatrix} dt \\
&= \int_a^b Fdl + i\int_a^b Gdl
\end{aligned}$$

And by Green's theorem we have:

$$= \int_{iut\Gamma} \mathrm{curl}F dxdy + i\int_{iut\Gamma} \mathrm{curl}G dxdy$$

And if we compute the curl:

$$\begin{aligned}
\mathrm{curl}F &= \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} = -\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \\
\mathrm{curl}F &= \frac{\partial G_2}{\partial x_1} - \frac{\partial G_1}{\partial x_2} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}
\end{aligned}$$

And by the Cauchy Riemann equations both are equal to 0. Hence:

$$\int_{\Gamma} f(z)dz = 0$$

Notice here we used that since u is simply connected and $\Gamma \subseteq u$ we have int $\Gamma \subseteq u$, so F, G are defined on int Γ

Example Let Γ be the unit circle, $f(z) = z^2$. By applying Cauchy theorem to $u = \mathbb{C}$, then:

$$\int_{\Gamma} f(z)dz = 0$$

Example ii Again, let's take Γ to be the unit circle and $f(z) = \frac{1}{z}$.

We are looking for a simply connected domain which contains the unit circle **and** f is holomorphic. But this is not possible because f is only holomorphic on $\mathbb{C}^*$ and there is no simply connected domain $U \subseteq \mathbb{C}^*$ that contains Γ. Therefore, Cauchy's theorem does not apply. (indeed, as we have calculated above, $\int_{\Gamma} f(z)dz \neq 0$)

Example iii Let $f(z) = \frac{1}{z}$, $\Gamma = \{z \in \mathbb{C} : \|z-2\| = 1\}$.

For instance let us take the domain $u = \{\mathrm{Re} > 0\}$ which is a simply connected domain which contains Γ, and where f is holomorphic on u. By the Cauchy theorem, we have that:

$$\int_{\Gamma} f(z)dz = 0$$

In practice, we'll apply Cauchy's theorem as follows:

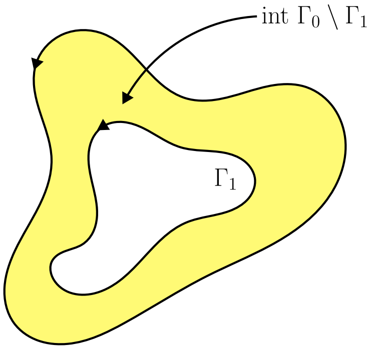
- Given Γ and f
- Find a simply connected open domain $U \subseteq \mathbb{C}$ such that:
  - U contains Γ and
  - f is holomorphic on U

There is an extension of Cauchy's theorem which is nice for application:

Theorem 8. Let Γ_0, Γ_1 be closed and simple regular curves such that $\Gamma_1 \subseteq \operatorname{int} \Gamma_0$. Let γ_0, γ_1 be parametrization of Γ_0, Γ_1 with the same orientation.

If f is holomorphic on our two curves, then:

$$\int_{\Gamma_0} f(z)dz = \int_{\Gamma_1} f(z)dz$$



This allows us to compute complicated line integral by an easy line integral.

How to shows this? Since f is holomorphic on int σ and intσ_2, by Cauchy's theorem:

$$\int_{\sigma_1} f(z)dz = 0 = \int_{\sigma_0} f(z)dz$$

So we have that:

$$\begin{aligned} 0 &= \int_{\sigma_0} f(z)dz + \int_{\sigma_1} f(z)dz \\ &= \int_{\gamma_0} f(z)dz - \int_{\gamma_1} f(z)dz + 0 \end{aligned}$$

Cauchy integral formula

Theorem 9. Suppose $D \subseteq \mathbb{C}$ is open and $f : D \to \mathbb{C}$ is holomorphic.

If $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz$$

For every closed, simple curve γ inside D whose interior contains z_0 and is counter-clockwise oriented.

How to show this? First:

$$\frac{f(z)}{z - z_0}$$

is holomorphic on $D \setminus \{z_0\}$. By extended Cauchy's theorem we have:

$$\int_{\gamma} \frac{f(z)}{z - z_0} = \int_{\gamma_1} \frac{f(z)}{z - z_0} dz$$

where $\gamma_1 : [0, 2\pi] \to \mathbb{C}$, $\gamma_1(t) = z_0 + re^{it}$, for a small r. Hence:

$$\begin{aligned}\int_\gamma \frac{f(z)}{z-z_0}dz &= \int_{\gamma_1} \frac{f(z)}{z-z_0}dz \\ &= \int_0^{2\pi} \frac{f\left(z_0 + re^{it}\right)}{re^{it}} ire^{it}dt \\ &= i\int_0^{2\pi} f\left(z_0 + re^{it}\right)dt \to i2\pi f(z_0) \text{ as } r \to 0\end{aligned}$$

using that f is continuous at z_0.

3.1.2 Trick to compute line integrals easily

- Apply Cauchy theorem $\to$ find simply connex set u 'enclosing' loop γ on which f is holomorphic
- Apply extension of the Cauchy theorem $\to$ replace $\int_{\gamma_0} fdz$ over a 'very complicated' Γ_0 by $\int_{\gamma_1} fdz$ over a 'simpler' Γ_1
- If Γ_0, Γ_1 have some endpoint and orientation and f is holomorphic on the region they enclose. Then:

$$\int_{\Gamma_0} fdz = \int_{\Gamma_1} fdz$$

Extension of Cauchy's integral formula

Under the same hypothesis as for the Cauchy's integral formula:

Theorem 10.

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^{(n+1)}}dz$$

Remark And we can see that if $n = 0$ then this fall under the usual formula.

This gives us another way to compute a maybe 'complicated' using something that 'may' be easier.

This essentially follows by differentiating the Cauchy's integral formula

Proof. *We first start from the Cauchy integral Formula:*

$$f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-z_0}dz$$

We can take the derivative on both side

$$f'(z) = \frac{1}{2\pi i}\frac{d}{dz_0}\int_\gamma \frac{f(z)}{z-z_0}dz$$

Using Leibniz rule, we move the derivative inside the integral:

$$f'(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{\partial}{\partial z_0}\left(\frac{f(z)}{z-z_0}\right)dz$$

Since $f(z)$ does not depend on z_0, only the denominator is differentiated, which means that we need to compute:

$$\frac{df}{dz_0}(z-z_0)^{-1} = (z-z_0)^{-2}$$

Now if we substitutes this into our equation

$$f'(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^2}$$

Now to prove the general result (not only the base case) we will need to do the induction steps. Assume that $f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^{n+1}}dz$. *If we retake its derivative as before we have*

$$\begin{aligned} f^{(n+1)}(z_0) &= \frac{n!}{2\pi i}\frac{d}{dz_0}\int_\gamma \frac{f(z)}{(z-z_0)^{n+1}}dz \\ &= \frac{n!}{2\pi i}\int_\gamma \frac{\partial}{\partial z_0}\frac{f(z)}{(z-z_0)^{n+1}} \end{aligned}$$

Let us now compute:

$$\frac{\partial}{\partial z_0}\frac{1}{(z-z_0)^{n+1}} = --\frac{n+1}{(z-z_0)^{n+2}}$$

which gives us:

$$f^{(n+1)}(z_0) = \frac{(n+1)!}{2\pi i}\int_\gamma \frac{f(z)}{(z-z_0)^{(n+1)+1}}dz$$

□

Remark 13. This formula is used to prove a statement that we have already seen: holomorphic functions are infinitely many times differentiable.

Example

Let $f(z) = \frac{\cos(z)}{z}$ and let γ be a simple closed curve, oriented counter-clockwise. The question that we are trying to answer is: what are the value of $\int_\gamma f dz$?

Before starting anything we can see our function kind look like what is being integrated above. Knowing that we can first check some information about our function:

- If $0 \in \gamma$ this integral is ill-defined
- If $0 \notin \gamma$ and 0 is not in the interior of γ then we have that our function is holomorphic even inside the curve. Therefore by Cauchy integral theorem:

$$\int_\Gamma f dz = 0$$

- If $0 \notin \gamma$ yet 0 is in the interior of γ then we note

$$\int_\gamma f(z)dz = \int_\gamma \frac{g(z)}{z}dz$$

By the Cauchy integral formula for $g(z) = \cos(z)$ and $z_0 = 0$ this gives us

$$= 2\pi i g(0) = 2\pi i$$

Example II

Let γ be the circle $\{z \in \mathbb{C} : \|z-2\| = 1\}$ oriented counter-clockwise. Compute $\int_\gamma \frac{e^z}{(z-2)^3}dz$.

This is a nice function under a power... First we can notice that e^z is holomorphic on $\mathbb{C}$ therefore, let us developed using the formula

$$= \int_\gamma \frac{f(z)}{(z-2)^{2+1}} = \frac{2\pi i}{2!}f(2) = \pi i e^2$$

Chapter 4

Laurent Series

Before jumping to Laurent series, first let's recall what Taylor series are and why are we interested in them.

4.1 Taylor series

The goal of Taylor series is to approximate any function using polynomial. The reason for this is that a polynomial is very nice to manipulate (derivative/primitive are very easy to compute) they are continuous, smooth, etc...

Definition 19. Taylor Series Let $f : \mathbb{R} \to \mathbb{R}$ sufficiently regular function. For any $x_0 \in \mathbb{R}$ the **Taylor polynomial** of degree N is

$$T_{x_0}^N f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(n)(x_0)}}{N!}(x - x_0)^N$$

Theorem 11. If $f \in C^{N+1}(\mathbb{R})$ then for all $x_1, x_0 \in \mathbb{R}$ there exists $\theta \in (0, 1)$ such that

$$f(x) = T_{x_0}^N f(x) + \frac{f^{N+1}(\theta x_0 + (1 - \theta)x)}{(N + 1)!}(x - x_0)^{N+1}$$

In practice, θ is not known, but one can usually estimate the $N + 1$ derivative. Also, Taylor's theorem informally means that we can approximate function by polynomials.

Definition 20. If f has derivatives of all orders, then we may consider the **Taylor series**

$$\sum_{n=0}^{\infty} \frac{f^{(n)(x_0)}}{n!}(x - x_0)^n$$

An important question which comes up using series is: Does this series actually converge? If we take for instance $f(x) = e^x, x_0 = 0$. The Taylor series around x_0 is given as

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

In this case is converges everywhere on $\mathbb{R}$ by definition of the exponential function.

A bad example of convergence this time would be $f(x) = \frac{1}{1-x}, x = 0$. This Taylor series around x_0 is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}}{n!}(x-0)^n = \sum_{n=0}^{\infty} x^n$$

since $f^{(n)}(x) = (1-x)^{-n-1}$.

This series on the right hand side converges for $\|x\| < 1$.

Remark 14. Functions f such that the Taylor series converges to f in a neighbourhood of x_0 are called **analytic**. Not all smooth function are analytic, e.g.

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

is smooth but not analytic.

4.2 Taylor series/ Regular Laurent Series for complex functions

Taylor series or more specifically regular Laurent series (which we will see later in this course) can approximate arbitrary precisely a function given a certain disk.

Theorem 12. Let $D \subseteq \mathbb{C}$ be open and $z_0 \in D$. Let $R > 0$ be such that the disk

$$B_R(z_0) := \{z \in \mathbb{C} : \|z - z_0\| < R\}$$

Is contained in D.

If $f : D \to \mathbb{C}$ is holomorphic then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)(z_0)}}{n!}(z - z_0)^n$$

For all $z \in B_r(z_0)$

What this means is that if I gave you a point and a ball around that point. Then, at each point around this ball the function at this point is equal to his Taylor series.

The largest possible R is called the **radius of convergence**

Remarks

1. In particular, all holomorphic functions are analytic
2. In practice: R is the distance of z_0 from the nearest '*point*' where f is not holomorphic/defined
3. We can connect the Taylor series to Cauchy's integral formula. This would say that f can be written as a power series

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}}{n!}(z - z_0)^n = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

Where

$$c_n = \frac{f^{(n)(z_0)}}{n!} = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{-z_0} d\zeta$$

For every simple, closed curve γ around z_0 with int $\gamma \subseteq D$ and counter-clockwise

Example Let $f(z) = e^z$ is holomorphic on $\mathbb{C}$. Therefore for every $z_0 \in \mathbb{C}$ the theorem applies for any $R > 0$ (so f has infinite radius of convergence) and

$$e^z = \sum_{n=0}^{\infty} \frac{e^{z_0}}{n!}(z - z_0)^n$$

Example II Let $f(z) = \frac{1}{1-z}$ holomorphic on $D = \mathbb{C} \setminus \{1\}$. Let $z_0 = 0$. In this case, the largest possible R is 1. For $z \in B_1(0)$ this gives us

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = \sum_{n=0}^{\infty} z^n$$

Since $f^n(0) = n!$

Example III Let $f(z) = \frac{1}{1+z^2}$ holomorphic on $D = \mathbb{C} \setminus \{-i, i\}$. Let $z_0 = 0$ which again give us as the largest R, 1. Using $B_1(0)$, we could simply use

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n$$

And insert $w = -z^2$ to get:

$$\frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}$$

Caution

The Taylor series representation around a given point is unique.

In particular, in order to compute the Taylor series representation at a given point, we don't have to do the 'straightforward' computation, we can use trick by using other Taylor series that we already know. This is something that we have already done in real analysis (for instance when doing limit of function of the type $e^{\sin(x)}$)

4.3 Laurent Series

The intuition behind Laurent series is that we can change our series in a way so that we can represent function even when the Taylor series is not defined. To do so we will allows our series to have **negative** exponent.

Definition 21. Let $D \subseteq \mathbb{C}$ be open and $z_0 \in D$. Assume $f : D \setminus \{z_0\} \to \mathbb{C}$ is holomorphic. Let $R > 0$ such that $B_R(z_0) \subseteq D$ Then for all $z \in B_R(z_0) \setminus \{z_0\}$

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

Where for every simple closed, counter-clockwise oriented curve γ such that $\operatorname{int} \gamma \subseteq D$ and z_0 is enclosed by γ,

$$c_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$$

Intuition The Laurent Series is like a Taylor series but now we include terms with polynomial singularities.

And as for the Taylor series, the Laurent series representation is **unique**.

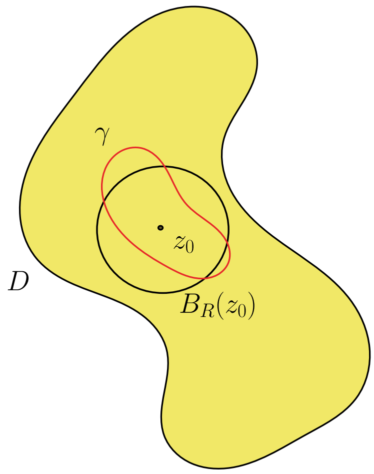


Figure 4.1: representation

If we find the coefficients c_n by any means (not necessarily by a path integral), then this has to be the Laurent series.

Definition 22.
Any Laurent Series splits into a **singular** and **regular** part

$$\sum_{n=-\infty}^{\infty} c_n(z-z_0)^n = \underbrace{\cdots + \frac{c-2}{(z-z_0)^2} + \frac{c-1}{z-z_0}}_{\text{singular part}} + \underbrace{c_0 + c_1(z-z_0) + \cdots}_{\text{regular part}}$$

In other words, the Taylor series is a Laurent series without singular part.

Example If $f : D \to \mathbb{C}$ holomorphic and $z_0 \in D$ then the Taylor series implies that

$$f(z) = \sum_{n=0}^{\infty} x_n(z-z_0)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

The singular part of the Laurent series has to vanish since for all $n \leq -1$ we have

$$\frac{f(\zeta)}{(\zeta-z_0)^{n!1}} \text{ is holomorphic near } z_0$$

By Cauchy's integral formula $\int_\gamma \frac{f(\zeta)}{(\zeta-z_0)^{n+1}} d = 0$

Example II For the following function $f : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ let us find the Laurent series around $z_0 = 0$

1. $f(z) = \frac{1}{z}$ This is already a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} c_n z^n \qquad \begin{cases} 1 & n = -1 \\ 0 & \text{otherwise} \end{cases}$$

2. $f(z) = \frac{5z+1}{z^2} = \frac{1}{z^2} + \frac{5}{z}$ which we can again translate into a Laurent series:

$$= \sum_{n=-\infty}^{\infty} c_n z^n$$

Where

$$c_n = \begin{cases} 1 & n = -2 \\ 5 = n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Remark 15. The Laurent series is unique. Hence, if we manage to compute a power series expansion for f as above by any means, (e.g. from another Laurent series), we have determined the Laurent series.

Example If we know that:

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad \text{for } |z| < 1$$

Then if we want to determine the Laurent series of $\frac{1}{1+z^2}$ we could have:

$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} \underbrace{=}_{|1|<1} \sum_{n=0}^{\infty} (-z^2)^n$$
$$= \sum_{n=0}^{\infty} c_k z^k$$

Where

$$c_k = \begin{cases} 0 & k \text{ odd} \\ (-1)^{\frac{k}{2}} & k \text{ even} \end{cases}$$

Example II Let $f : \mathbb{C} \setminus \{0, -1\} \to \mathbb{C}$, $f(z) := \frac{1}{z(z+1)}$. What we are trying to do is to develop the Laurent series of f at a given point $z_0 \in \mathbb{C}$. In practice, the way to go is to look at the function, to see where it is behaving nicely or not. But here as we don't know (for an arbitrary point z_0) we need three cases:

1. Assume $z_0 = 0$, we can observe that $f(z) = \frac{1}{z} - \frac{1}{z+1}$. But here we just summoned a known Taylor series $\frac{1}{z-(-1)}$ Which gives us

$$\frac{1}{1-(-z)} = \sum_{n=0}^{\infty} (-1)^n z^n \quad \text{where } |z| < 1$$

So $f(z) = \frac{1}{z} - \sum_{n=0}^{\infty} (-1)^n z^n$ But here the $\frac{1}{z}$ is just a term of the regular part of the Laurent series, therefore this gives us:

$$= \sum_{n=-1}^{\infty} (-1)^{n+1} z^n$$

The radius of convergence for this series is give by $R = 1$, it is the distance to z_0 to the next distinct singular point of f(which is given by $z = -1$).

2. Assume $z_0 = -1$. Using the same trick as in (1): $f(z) = -\frac{1}{z-(-1)} + \frac{1}{z}$. Now if we expand $\frac{1}{z}$ as Taylor series around -1:

$$\frac{1}{z} = \sum_{n=0}^{\infty} (-1)(z-(-1))^n \quad \text{where } |z-(-1)| < 1$$

so

$$f(z) = -\frac{1}{z-(-1)} + \sum_{n=0}^{\infty} (-1)(z-(-1))^n$$
$$= \sum_{n=-1}^{\infty} (-1)(z-(-1))^n$$

With a Radius of converge of 1

3. Assume now that $z_0 \neq 0, -1$. In this case, f is holomorphic on an open set containing z_0 so around z_0 f can be written as a regular Taylor series. This

series converges in any disk around z_0 which does not contain 0 or -1. The radius is given by $R = \min\{|z_0|, |z_0+1|\}$ Does are the distance between our point z_0 and the point 0 and -1.

The principle is to use as much knowledge as we know about Taylor/Laurent series and common functions in order to compute new series.

4.3.1 Regular point

Regular point

Definition 23. Regular point

1. We say f has a **regular point** at z_0 if the singular part of the Laurent series at z_0 vanishes
2. We say f has a **pole of order m** at z_0 if the coefficients of the Laurent series of f at z_0 satisfy $c_n = 0 \forall n < -m$, i.e.

$$f(z) = \frac{c_m}{(z-z_0)^m} + \frac{c_{m+1}}{(z-z_0)^{m-1}} + \cdots, \quad c_m \neq 0$$

3. We say f has an essential singularity at z_0 if its Laurent series has infinitely many coefficients $c_n \neq 0$ with $n \leq 1$

The intuition behind the second point if that '*f blows up like $\frac{1}{(z-z_0)^m}$ at z_0*'

Example Let $f(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z$, here 0 is regular.

Remark If f is holomorphic at z_0, then z_0 is a regular point. The converse is also true: if z_0 is a regular point of f, then

$$\lim_{z \to z_0} f(z)$$

Exists and if we extend f to z_0 this way, f is holomorphic at z_0. In short

$$z_0 \text{ is regular point of } f \iff \lim_{z \to z_0} f(z) \text{ exists}$$

Example II

- Let $f(z) = \frac{1}{z}$ has a pole of order 1 at $z_0 = 0$
- $f(z) = \frac{1}{(z+2)^2} + \frac{1}{z}$ has a pole of order 2 at $z_0 = -2$ and a pole of order 1 at $z_0 = 0$

The second statement comes from the fact that the function can be directly written as a Laurent series with the squared represent the second part of the Laurent series which implies an order of 2 this part will only contribute to the Taylor series (because this function is holomorphic). $\frac{1}{z}$ will definitely belongs to the singular part.

If f has a pole of order m at z_0, then we can write $f(z)$ to be:

$$f(z) = \frac{c_{-m}}{(z-z_0)^m} + \frac{c_{-m+1}}{(z-z_0)^{m-1}} + \cdots$$

Which means that if we take for instance $g(z) = (z-z_0)^m f(z)$ has a regular point at z_0

In fact f has a pole of order m at z_0 if and only if $\lim_{z \to z_0}(z-z_0)^m f(z)$ exists and is non-zero.

Which means that we can multiply our function by a power of z and the function will still exists (which is kind of obvious) so for instance $x^{2000} \cdot \frac{1}{z}$ when going to 0 the limit will exists and will be 0 (because the order is 1 and not 2000). In particular if f has a pole of order m at z_0, we can write

$$f(z) = \frac{g(z)}{(z-z_0)^m}$$

where g is holomorphic at z_0 and $g(z_0) \neq 0$ (the opposite is also true)

Remark 16. The fact that we can rewrite f as a product $\frac{g(z)}{z-z_0}$ for a given singularity is pretty big. Using Cauchy's Integral Formula, this implies that if $z_0 \in \gamma$ and that $g(z)$ is analytic inside and on γ, then:

$$\oint_\gamma \frac{g(z)}{z-z_0} dz = 2\pi i \cdot g(z_0)$$

This means that we can solve compute integral of another function to get the integral of our function f.

Example III Let us for instance consider $f(z) = e^{\frac{1}{z}}$ at $z_0 = 0$. We can use the Taylor series of the exponential in order to compute the Laurent series:

$$f(z) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{z}\right)^n}{n!} = \sum_{n=-\infty}^{0} \frac{1}{(-n)!} z^n$$

So 0 is an essential singularity of f.

Example III ii Let $f(z) = ze^{\frac{1}{z}}$ at $z_0 = 0$. Using the same trick as above

$$f(z) = z \sum_{n=-\infty}^{0} \frac{1}{(-n)!} z^n = \underbrace{z+1}_{\text{regular part}} + \underbrace{\frac{1}{2!z} + \frac{1}{3!z^2} + \cdots}_{\text{singular part}}$$

So 0 is an essential singularity

Now let's discuss quotients of Taylor series in this context. Suppose $p, q : D \setminus \{z_0\} \to \mathbb{C}$ holomorphic with z_0 being regular for both function. Regular means that we can write the Laurent series as:

$$p(z) = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots$$

$$q(z) = b_0 + b_1(z-z_0) + b_2(z-z_0)^2 + \cdots$$

The goal is to study singularities of the quotients $\frac{p}{q}$ near z_0. In particular we are asking ourself what is the order of the pole (if there is any)? Let's discuss a few instructive examples. What we will harm here is our q, we will need to be careful when $q(z) = 0$, we will need to do some assumptions:

1. if $b_0 \neq 0$ so $q(z_0) \neq 0$. In particular $\frac{1}{q}$ is holomorphic near z_0 and we have that

$$\lim_{z\to z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)} = \frac{a_0}{b_0}$$

   exists which implies that z_0 is a regular point for $\frac{p}{q}$
2. If $a_0 = b_0 = 0$ and this case both function vanish at 0... To make our life easier let us assume that $b_1 \neq 0$ then

$$\begin{aligned}\lim_{z\to z_0} \frac{p(z)}{q(z)} &= \lim_{z\to z_0} \frac{a_1(z-z_0) + a_2(z-z_0)^2 + \cdots}{b_1(z-z_0) + b_2(z-z_0)^2 + \cdots} \\ &= \lim_{z\to z_0} \frac{a_1 + a_2(z-z_0) + \cdots}{b_1 + b_2(z-z_0) + \cdots} = \frac{a_1}{b_1}\end{aligned}$$

because the limit exists, z_0 is a regular point for f.

Similarly if $a_0 = b_0 = 0$ and $a_1 = b_1 = 0$ yet $b_2 \neq 0$.

Example Let our function be $f(z) = \frac{z}{\sin z}$ where $z_0 = 0$. Using power series expansion of the sin function, we can convince ourself that the limit of z approaching 0 does exists (recall analysis I):

$$f(z) = \frac{z}{z - \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots} = \frac{1}{1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots}$$

By setting $z = 0$ we can see that the limit does exists:

$$\lim_{z \to 0} f(z) = 1$$

Therefore, 0 is a regular point

Summary In Summary, in order to see if the limit exists, either expand in power series and take out as many common powers of $z - z_0$ from the nominator and denominator as possible and take the limit, or use l'Hospital's rule.

Remark 17. This looks a lot like how we used to compute limit in analysis I and yes it is, We use the same principle here than there.

Example Suppose that

$$p(z) = a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \cdots, \quad a_k \neq 0$$
$$q(z) = b_l(z - z_0)^l + b_{l+1}(z - z_0)^{l+1} + \cdots, \quad b_l \neq 0$$

Consider now the function $f(z) = \frac{p(z)}{q(z)}$ near z_0.

- If $k \geq l$ then z_0 is a regular point for f. To do so we will show that the limit of our function when approaching the point converges

$$\lim_{z \to z_0} f(z) = \lim_{z \to z_0} \frac{a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \cdots}{b_l(z - z_0)^l + b_{l+1}(z - z_0)^{l+1} + \cdots}$$

  Because $k \geq l$, we can factor out $(z - z_0)^l$ which gives us

$$\begin{aligned} &= \lim_{z \to z_0} \frac{a_k(z - z_0)^{k-l} + a_{k+1}(z - z_0)^{k+1-l} + \cdots}{b_l + b_{l+1}(z - z_0) + \cdots} \\ &= \begin{cases} 0 & \text{if } k > l \\ \frac{a_k}{b_l} & \text{if } k = l \end{cases} \end{aligned}$$

- If $k < l$ then z_0 is a pole of order $l - k$. Using the same trick as what we have done above, factoring out until we see holomorphic function:

$$\begin{aligned} f(z) &= \frac{a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \cdots}{b_l(z - z_0)^l + b_{l+1}(z - z_0)^{l+1} + \cdots} \\ &= \frac{a_k + a_{k+1(z-z_0)+\cdots}}{b_l(z - z_0)^{l-k} + b_{l+1}(z - z_0)^{l+1-k} + \cdots} \\ &= \underbrace{\frac{1}{(z - z_0)^{l-k}}}_{\text{blows up at } z_0} \left[\frac{a_k + a_{k+1}(z - z_0) + \cdots}{b_l + b_{l+1}(z - z_0) + \cdots} \right] \end{aligned}$$

  This right part is holomorphic at z_0 Laurent series with only regular part. Therefore, we could write our function:

$$f(z) = \frac{c_0}{(z-z_0)^{l-k}} + \frac{c_1}{(z-z_0)^{l-k+1}} + \cdots$$

Example Let $f(z) = \frac{\sin z}{2z^3+z^4}$ has a pole of order 2 at $z_0 = 0$.

Since $\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$ we have that

$$f(z) = \frac{z - \frac{z^3}{z!} + \frac{z^5}{5!} - \cdots}{2z^3+z^4} = \frac{1 - \frac{z^2}{3!} + \frac{z^4}{5!} + \cdots}{2z^2+z^3}$$
$$= \frac{1}{z^2}\left[\frac{1 - \frac{z^2}{3!} + \frac{z^4}{5!}}{2+z}\right]$$

Example Let $f(z) = z\sin\left(\frac{1}{z}\right)$, the claim is that this function has an essential singularity at 0.

Indeed, we can realise this by using the Taylor expansion for this sin function which will gives us a Laurent series:

$$\sin\frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^k}{(2n+1)!}\left(\frac{1}{z}\right)^{n+1}$$

And hence, by multiplying this sum by z:

$$z\sin\frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}\frac{1}{z^{2n}}$$

And this indeed converges when $z \neq 0$.

4.4 Residues Theorem

A residue is a very simple object: it is the first coefficient of the Laurent series

Definition 24. Residue The coefficient c_{-1} of the term $(z-z_0)^{-1}$ in the Laurent series for f at z_0 is called the **residue** of f at z_0. Denoted by:

$$\mathrm{Res}_{z_0}(f) := c_{-1}$$

Remark 18. We have already seen that the coefficients of the Laurent series can be described directly using line integral.

$$\mathrm{Res}_{z_0}(f) = \frac{1}{2\pi i}\int_\gamma f(z)dz$$

Where γ is a closed, simple, counterclockwise oriented curve that contains z_0 in its interior and f is holomorphic on the int $\gamma \setminus \{z_0\}$

Now let us compute some residue!!!!!!!!!!!!!!!!!!

Example If f is holomorphic on an open set containing z_0,

$$\mathrm{Res}_{z_0}(f) = 0$$

By Cauchy's theorem

The concept of residue becomes interesting when we have singularity inside our function, especially at 0. For instance if we take:

$$\mathrm{Res}_{z_0}\left(\frac{1}{z}\right)$$

Then we can see that we already have the Laurent series of $\frac{1}{z}$ at the point 0.

$$\mathrm{Res}_{z_0}\left(\frac{1}{z}\right) = 1$$

Suppose more generally f has a pole of order m at z_0, i.e. its Laurent series reads

$$f(z) = c_{-m}(z - z_0)^{-m} + \cdots + c_{-1}(z - z_0)^{-1} + c_0 + \cdots$$

Where $c_{-m} \neq 0$, consider the holomorphic function

$$\begin{aligned} g(z) &= (z - z_0)^m f(z) \\ &= c_{-m} + c_{-m+1}(z - z_0) + \cdots + c_{-1}(z - z_0)^{m-1} + \cdots \end{aligned}$$

Then we can extract the residue by the formula

$$\begin{aligned} c_{-1} &= \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}}\Big|_{z_0}(z) \\ &= \lim_{z \to z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}}((z - z_0)^m f(z)) \end{aligned}$$

This formula is useful if we already know that f has a pole of order m at z_0. In particular if $m = 1$ then this becomes

$$\mathrm{Res}_{z_0}(f) = \lim_{z \to z_0} (z - z_0) f(z)$$

Example Let us consider $f(z) = \frac{1}{\sin z}$ at $z_0 = 0$ which has a pole of order 1:

$$f(z) = \left[z - \frac{z^3}{3!} + \cdots\right]^{-1} = \frac{1}{z} \underbrace{\left[1 - \frac{z^2}{3!} + \cdots\right]^{-1}}_{\text{regular}}$$

Hence from the above example,

$$\begin{aligned} \mathrm{Res}_0(f) &= \lim_{z \to 0} z f(z) \\ &= \lim_{z \to 0} \frac{z}{\sin z} \\ &= \lim_{z \to 0} z \left[z - \frac{z^3}{3!} + \cdots\right]^{-1} \\ &= \lim_{z \to 0} \left[1 - \frac{z^2}{3!} + \cdots\right]^{-1} = 1 \end{aligned}$$

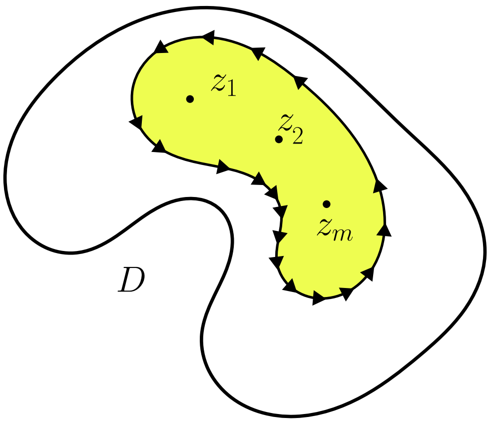
4.4.1 Residue Theorem

Theorem 13. Suppose $D \subseteq \mathbb{C}$ is an open and simply connected set and let γ be a simple closed curve in D which is counterclockwise oriented. Let $z_1, \ldots, z_m \in \text{ int}\gamma$.

Assume $f : D \setminus \{z_1, \ldots, z_m\} \to \mathbb{C}$ is holomorphic

then

$$\int_\gamma f(z)dz = 2\pi i[\text{Res}_{z_1}(f) + \cdots + \text{Res}_{z_m}(f)]$$

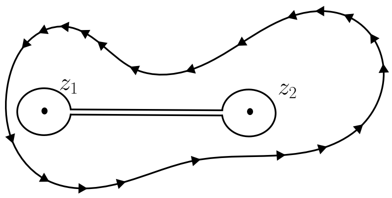


Hence, in order to compute the line integral of $\int_\gamma f dz$ it suffices to compute the residues of the singular points in the interior of γ.

How does one show this?

For simplicity, we will consider for 2 points only (the same argument foes for n points)

What we are doing geometrically is creating two curves ρ_1, ρ_2 that turns around z_1 and z_2 and join them together with a very small line, geometrically this gives us something like this:



By the extension of Cauchy's theorem

$$\int_\gamma f dz = \int_\rho f dz$$

Then we are trying to see what happens with the integral of ρ when the radius becomes very thin. First, for the '*bridge*', as it becomes very narrow, it will cancel out. This leaves us with the integral over ρ_1, ρ_2. Therefore

$$\int_\gamma f dz = \int_{\rho_1} f dz + \int_{\rho_2} f dz = 2\pi i\text{Res}_{z_1}(f) + 2\pi i\text{Res}_{z_2}(f)$$

Remark Recall that:

$$\int_{\rho_1} f dz = 2\pi i\text{Res}_{z_1}(f)$$

Comes down from the definition of the Residue seen in the Remark 18.

Example Consider $f(z) = \frac{1}{z}$. Let γ be a simple closed counterclockwise curve in $\mathbb{C}$. Compute $\int_\gamma f dz$

1. If $0 \in$ int γ then by the residue theorem
$$\int_\gamma f dz = 2\pi i Res_0(f) = 2\pi i$$
2. If $0 \notin$ int $\gamma \cup \gamma$ then
$$\int_\gamma f dz = 0$$
3. if $0 \in \gamma$ then
$$\int_\gamma f dz \text{ is not well-defined}$$

Example Let $f : \mathbb{C} \setminus \{0, -1\} \to \mathbb{C}$, $f(z) = \frac{1}{z(z+1)}$. We have already seen f has two poles of order 1 at $z_0 = 0$ and $z_1 = -1$. Hence

Let γ be a closed counterclockwise oriented curve. The queis always the same: what is $\int_\gamma f dz$?

1. 0, $-1 \notin$ int $\gamma \cup \gamma$ then $\int_\gamma f dz = 0$
2. $0 \in$ intγ but $-1 \notin$ int $\gamma \cup \gamma$ then
$$\int_\gamma f dz = 2\pi i \mathrm{Res}_0(f) = 2\pi i$$
3. $0 \notin$int $\gamma \cup \gamma$ but $-1 \in$ int γ then
$$\int_\gamma f dz = 2\pi i \mathrm{Res}_{-1}(f) = -2\pi i$$
4. If $0, -1 \in$ int γ then by the residue theorem
$$\int_\gamma f dz = 2\pi i[\mathrm{Res}_0(f) + \mathrm{Res}_{-1}(f)] = 0$$
5. $0 \in \gamma$ or $-1 \in \gamma$ then the integral is not well defined

What is important is that when we are asked to compute simple line integral, it is important the divide cases based the singularities inside of γ

Example Let $f : \mathbb{C} \setminus \{0, 1\} \to \mathbb{C}$, $f(z) = \frac{1}{z^2} + \frac{2}{z} + \frac{3}{z-1}$. This function has a pole of order 2 at 0 and a pole of order 1 at 1

Example Let $\exp\left(\frac{1}{z^2}\right) = 1 = \frac{1}{z^2} + 1\frac{1}{2!}\frac{1}{z^2} + \cdots$ hence
$$\mathrm{Res}_0\left(\exp\left(\frac{1}{z^2}\right)\right) = 0$$

4.4.2 Applications

The residue theorem can be used to compute integrals over simple closed curve, sometimes even to compute **real** integral by replacing them with complex integrals.

Example Let us for instance compute

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

If you have forgotten that this is the primitive of arctan then you can still try to rephrase this into a complex integral. Let us set $f(z) = \frac{1}{1+z^2}$.

Given $R > 0$, consider the segment $L_R : [-R, R] \to \mathbb{C}$ given by $L_R(t) = t$ and the half circle $c_R : [0, \pi] \to \mathbb{C}$ given by $C_R(t) = Re^{it}$

Consider the closed curve $\gamma = LR \cup C_R$. Then the function $f(z) = \frac{1}{1+z^2}$ has poles of order 1 at i. From the residue theorem then

$$\int_\gamma f dz = 2\pi i \mathrm{Res}_i(f)$$

But we can compute the residue explicitly which is given by

$$\mathrm{Res}_i(f) = \lim_{z \to i} (z - i) f(z) = \lim_{z \to i} \frac{1}{z+i} = \frac{1}{2i}$$

So:

$$\begin{aligned} \int_\gamma d dz = \pi &= \int_{L_R} f dz + \int_{C_R} f dz \\ &= \int_{-R}^{R} \frac{1}{1+t^2} dt \underbrace{\to}_{R \to \infty} \int_{-\infty}^{\infty} \frac{1}{1+t^2} dt \end{aligned}$$

On the other hand,

$$\int_{C_R} f dz = \int_0^\pi \frac{1}{1+R^2 e^{2it}} Rie^{it} dt \approx \frac{\pi}{R} \underbrace{\to}_{R \to \infty}$$

This method works for any integrand that decays faster than $\frac{1}{R}$ as $R \to \infty$

Example Let $R(x) = \frac{P(x)}{Q(x)}$ where P, Q are real polynomials and $\deg Q \geq \deg P + 2$. Assume R is defined on $\mathbb{R}$ which means that $Q(x) \neq 0 \quad \forall x \in \mathbb{R}$

Observations

1. The latter condition on Q forces $\deg Q$ to be even
2. Q has real coefficients if $Q(z) = 0$ for $z \in \mathbb{C}$ then by taking the complex conjugate of this equations, we find $Q(\overline{z}) = 0$ So roots of Q in $\mathbb{C}$ come in pairs of complex conjugates (none of which are real)

Given $a \geq 0$ we want to compute

$$\int_{-\infty}^{\infty} R(x) \cos(ax) dx, \quad \int_{-\infty}^{\infty} R(x) \sin(ax) dx$$

By Euler's formula $e^{iax} = \cos(ax) + i \sin(ax)$, we can compute both by computing

$$\int_{-\infty}^{\infty} R(x) e^{iax}$$

Remark And then we can read off the real and imaginary part of the result. Such integrals appear often in applications (e.g. Fourier transform, resonances in harmonic systems with friction, etc.).

Consider $f(z) = R(z)e^{iaz}$, let $z_1, \ldots, z_k$ be zeros of Q, then:

$$f : \mathbb{C} \setminus \{z_1, \ldots, z_k\} \to \mathbb{C}$$

is holomorphic.

The goal now will be to use the same trick as before, making our path be a half circle in the upper complex plane and a line on the Real axis, but this time we need a r large enough so that $z_1, \ldots, z_k$ are contained in the disk $B_r(0)$.

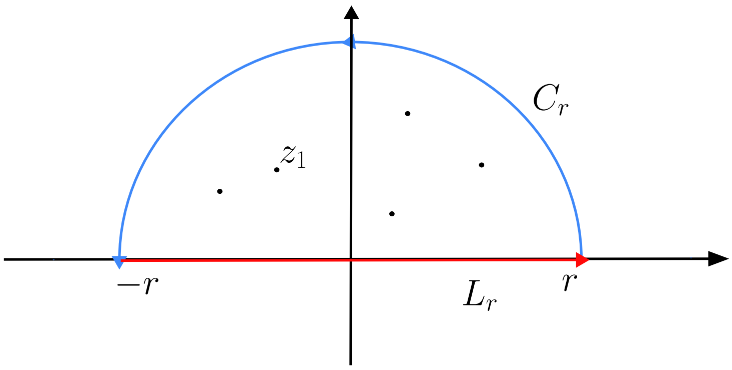


Figure 4.2: Path $C_r + L_r$

As we can see above, we can construct our two paths:

$$L_r = \{z \in \mathbb{C} : \text{Im } z = 0, -r \leq \text{Re } z \leq rz\}$$

parametrized by $L_r(t) = t$, $t \in [-r, r]$

$$C_r = \left\{re^{i\theta} : \theta \in [0, \pi]\right\}$$

Now we can defined the whole path:

$$\Gamma_r := L_R \bigcup C_r$$

and the integral over this path:

$$\int_{\Gamma_r} f dz = \underbrace{\int_{L_r} f dz}_{=\int_r^r R(t)e^{iat}dt} + \int_{C_r} f dz$$

As $r \to \infty$, one can compute $\int_{\Gamma_r} f dz$ by using the residue theorem: if $z_1, \ldots, z_k$ are zeros of Q in the upper half of the plane (which are half of the total number of zeroes of Q) then:

$$\int_{\Gamma_r} f dz = 2\pi i[\text{Res}_{z_1}(f) + \cdots + \text{Res}_{z_m}(f)]$$

As $r \to \infty$ this goes to $\int_{-\infty}^{\infty} R(x)e^{iax}dx$.

It remains to show that $\int_{C_r} f dz \to 0$ as $r \to \infty$.

- For Im $z \geq 0$ and $a \geq 0$, $\left|e^{iaz}\right| \leq 1$. Indeed, if we split z as $z = x + iy$ we can see:

$$\left|e^{ia(x+iy)}\right| = \left|e^{aix}e^{-ay}\right| = e^{-ay} \leq 1$$

Remark $\left|e^{ait}\right|$ is not bounded on all of $\mathbb{C}$, here it is because of the fact $i^2 < 0 \implies e^{\text{something} \leq 0} \leq 1$ But if $\operatorname{Im} z < 0$ then with have a positive exponent to our number $\implies$ explodes

- $|R(z)|$ decays to 0 as $|z| \to \infty$

$$\begin{aligned} \left|\int_{C_r} f dz\right| &= \left|\int_0^\pi f\big(re^{i\theta}\big)ire^{i\theta}d\theta\right| \\ &\leq \int_0^\pi \left|f\big(re^{i\theta}\big)\right|\left|rie^{i\theta}\right|d\theta \\ &= \int_0^\pi \frac{\left|P\big(re^{i\theta}\big)\right|}{\left|Q\big(re^{i\theta}\big)\right|}\left|e^{iare^{i\theta}}\right|rd\theta \end{aligned}$$

The first inequality comes from the triangle inequality. Then using the same fact as on the first point, $\left|e^{iare^{i\theta}}\right| \leq 1$ since $\operatorname{Im}\big(re^{i\theta}\big) \geq 0$ for $\theta \in [0, \pi]$

To see where the part involving $\frac{P}{Q}$ tends to, not if

$$\left.\begin{aligned} P(z) &= b_l z^l + b_{l-1} z^{l_1} + \cdots \\ Q(z) &= a_k z^k + a_{k-1} z^{k-1} + \cdots \end{aligned}\right\} \frac{P(z)}{Q(z)} = \frac{1}{z^{k-l}} \frac{b_l + \frac{b_{l-1}}{z} + \cdots}{a_k + \frac{a_{k-1}}{z} + \cdots}$$

Hence

$$\left|\frac{P\big(re^{i\theta}\big)}{Q\big(re^{i\theta}\big)}\right| \leq \frac{\text{Const}}{r^{k-l}} \text{ for large } r$$

And thus:

$$\left|\int_{C_r} f dz\right| \leq \frac{\text{const}\pi}{r^{k-l}} r \to 0 \text{ as } r \to \infty$$

In summary we deduce:

$$\boxed{\int_{-\infty}^{\infty} R(x)e^{iax}dx = 2\pi i[\operatorname{Res}_{z_1}(f) + \cdots + \operatorname{Res}_{z_m}(f)]}$$

Where $z_1, \ldots, z_m$ are zeros of Q in the upper half of the plane and $f(z) = R(z)e^{iaz}$

Benefit One can compute the residues of f by finding the poles of it and their orders, without computing Laurent series!

Example We want to compute the integral:

$$\int_{-\infty}^{\infty} \frac{x^2}{16 + x^4} \cos x dx$$

This is of the shape form the previous example with $a = 1$, $P(x) = z^2$, $Q(x) = 1 + x^4$, Thus the integral coincides with:

$$\int_{-\infty}^{\infty} \frac{x^2}{16 + x^4} \cos x dx = \operatorname{Re}\left(\int_{-\infty}^{\infty} \frac{x^2}{16 + x^4} e^{ix} dx\right)$$

The singularities of $f(z) = \frac{z^2}{16+z^4}$ correspond to the complex root of $16 + z^4 = 0 \implies r = 2, \theta = \frac{\pi}{4} + k\frac{\pi}{2}$ where $k \in \{0, \ldots, 3\}$

Therefore we have our four singularities:

$$z_1 = \sqrt{2} + \sqrt{2}i z_2 = -\sqrt{2} + \sqrt{2}i z_3 = -\sqrt{2} - \sqrt{2}i z_4 = \sqrt{2} - \sqrt{2}i$$

Hence we define Q as

$$Q(z) = (z - z_1)(z - z_2)(z - z_3)(z - z_4)$$

So

$$R(z) = \frac{P(z)}{Q(z)} = \frac{z^2}{(z-z_1)(z-z_2)(z-z_3)(z-z_4)}$$

Where each z_i is a pole of order 1.

Only the z_i with an Im ≥ 0 are relevant for the integral which implies that we only care for the Residue of z_1 and z_2:

$$\begin{aligned}\mathrm{Res}_{z_1}(f) &= \lim_{z\to z_1}(z-z_1)f(z) = \frac{e^{-\sqrt{2}}e^{\sqrt{2}i}}{4\sqrt{8}}\frac{1+i}{i}\\ \mathrm{Res}_{z_2}(f) &= \lim_{z\to z_2}(z-z_2)f(z) = \frac{e^{-\sqrt{2}}e^{-\sqrt{2}i}}{4\sqrt{8}}\frac{1-i}{i}\end{aligned}$$

Hence:

$$\begin{aligned}\mathrm{Re}\left(\int_{-\infty}^{\infty} R(x)e^{ix}dx\right) &= \mathrm{Re}(2\pi i(\mathrm{Res}_{z_1}(f) + \mathrm{Res}_{z_2}(f)))\\ &= 4\pi e^{-\sqrt{2}}\left(\cos\sqrt{2} - \sin\sqrt{2}\right)\end{aligned}$$

Integrals of rational function of trigonometric functions

Suppose $P(x,y), Q(x,y)$ are polynomials and Q is non-zero on the unit circle, i.e. $Q(x,y) \neq 0$ where $x^2+y^2=1$ we want to compute:

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)}dt$$

We can re express this as a complex integral over the unit circle. Recall

$$\begin{aligned}\cos t &= \frac{e^{it}+e^{-it}}{2}\\ \sin t &= \frac{e^{it}-e^{-it}}{2i}\end{aligned}$$

We have:

$$\begin{aligned}\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)}dt &= \int_0^{2\pi} \frac{P\left(\frac{e^{it}+e^{-it}}{2}, \frac{e^{it}-e^{-it}}{2j}\right)}{Q\left(\frac{e^{it}+e^{-it}}{2}, \frac{e^{it}-e^{-it}}{2i}\right)}dt\\ &= \int_0^{2\pi} \frac{Q(\cdots)}{Q(\cdots)}\frac{1}{ie^{it}}ie^{it}dt\end{aligned}$$

Now we can set

$$f(z) = \frac{P\left(\frac{z+\frac{1}{z}}{2}, \frac{z-\frac{1}{z}}{2i}\right)}{Q\left(\frac{z+\frac{1}{z}}{2}, \frac{z-\frac{1}{z}}{2i}\right)}\frac{1}{iz}$$

Then the integral in question is equal to:

$$\int_0^{2\pi} f\left(e^{it}\right)ie^{it}dt = \int_\gamma f dz$$

Where $\gamma : [0, 2\pi] \to \mathbb{C} \quad \gamma(t) = e^{it}$

Since P, Q are polynomials,f is holomorphic on $C \setminus \{z_1, \ldots, z_k\}$ where $z_1, \ldots, z_k$ are the point where the denominator vanishes

Remark A function which is holomorphic except for isolated points is called **meromorphic**

If $z_1, \ldots, z_l$ are the points of $z_1, \ldots, z_k$ lying in the **interior of the unit circle**, the reside theorem implies:

$$\boxed{\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} dt = 2\pi i [\mathrm{Res}_{z_1}(f) + \cdots + \mathrm{Res}_{z_l}(f)]}$$

Example Let us now try to compute $\int_0^{2\pi} \frac{1}{2 \cos\theta} d\theta$. This falls into the previous example with $P(x, y) = 1$, $Q(x, y) = 2 + x$. Computing the function f from the previous example explicitly yields

$$f(z) = \frac{2}{i} \frac{1}{z^2 + 4z + 1}$$

The singular points are on the zeros of $z^2 + 4z + 1$ which are $z_1 = -2 + \sqrt{3}$ and $z_2 = -2 - \sqrt{3}$. Only z_1 lies in the unit circle. Moreover f has a pole of order 1 at z_1. Hence

$$\mathrm{Res}_{z_1}(f) = \lim_{z \to z_1} (z - z_1) f(z) = \frac{1}{\sqrt{3} i}$$

Consequently,

$$\int_0^{2\pi} \frac{1}{2 + \cos\theta} d\theta = 2\pi i \mathrm{Res}_{z_1}(f) = \frac{2\pi}{\sqrt{3}}$$

Part III

Integral Transforms

Chapter 5

Review of Fourier Analysis

We will first do a first review of Fourier transform in order to have a better introduction to Laplace transform.

Fourier transform

Definition 25. A Fourier transform is a function $\hat{f} : \mathbb{R} \to \mathbb{C}$ of a given function $f : \mathbb{R} \to \mathbb{C}$, whenever f is well-defined, is given as:

$$\hat{f}(a) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x) dx$$

The *well-defined* here is a bit sloppy but it is wanted, the transform is defined when we are able to compute the integral written above. But if we want to be a bit more formal we will say the well-defined means where f is piecewise continuous and $\int_{-\infty}^{\infty} |f| dz < \infty$ (meaning, $f \in L^1(\mathbb{R}; \mathbb{C})$)

Remark We also note the transform as $\mathcal{F}[f]$ instead of $\hat{f}$.

Inverse Fourier transform

Definition 26. Inverse Fourier transform The inverse Fourier transform of a function $\hat{f} : \mathbb{R} \to \mathbb{C}$ is the function $f : \mathbb{R} \to \mathbb{C}$ defined as (whenever $\hat{f}$ is defined)

$$\check{f}(x) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iax} f(a) da$$

Remark It is also noted as $\mathcal{F}^{-1}[f]$ instead of $\check{f}$

Under the appropriate hypothesis of f, it is the inverse of the Fourier transform meaning:

$$\mathcal{F}^{-1}[\mathcal{F}[f]] = f$$

Remark 19. In general $\hat{f}$ is complex-valued even if f is real-valued. However, if f is real-valued and is even ($f(x) = f(-x) \quad \forall x \in \mathbb{R}$) then $\hat{f}$ is also real-valued.

Signal of "Physical domain"

The Fourier transform allows us given a signals to answer the question '*How much of every frequency is in the signal*'

Properties of Fourier transform

The Fourier transform has several properties:

1. Linearity: $\mathcal{F}[\lambda f + \mu g] = \lambda \mathcal{F}[f] + \mu \mathcal{F}[g] \quad \lambda, \mu \in \mathbb{R}$

2. Time shift: Let $g(x) = f(x - x_0)$ then

$$\begin{aligned}
\hat{g}(a) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x - x_0) dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x - x_0) dx \\
&\underbrace{=}_{y = x - x_0} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia(y + x_0)} f(y) dy \\
&= \frac{e^{-iax_0}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iay} f(y) dy \\
&= e^{-iax_0} \hat{f}(a)
\end{aligned}$$

3. Frequency shift: let $g(x) = e^{ia_0 x} f(x)$ for $a_0 \in \mathbb{R}$

$$\begin{aligned}
\hat{g}(a) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} g(x) dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} e^{ia_0 x} f(x) dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i(a - a_0)x} f(x) dx = \hat{f}(a - a_0)
\end{aligned}$$

This can be seen as an '*inverse*' of the second property

4. Rescaling / Dilation. Let $g(x) = f(\lambda x)$ where $\lambda \in \mathbb{R} \setminus \{0\}$

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(\lambda x) dx$$

To recover $\hat{f}$ we want to substitute $y = \lambda x$. However we have to be careful about this substitution – we have two cases, λ positive or negative. When λ is negative, it flips the orientation of the real line
- If $\lambda > 0$ we get $\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda} dy$
- If $\lambda < 0$ we get $\int_{\infty}^{-\infty} \frac{1}{\lambda} dy = -\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda} dy$ By merging the minus with the λ we get that this is equal to $= \int_{-\infty}^{\infty} \cdots \frac{1}{|\lambda|} dy$

In both cases we have:

$$\begin{aligned}
\hat{g}(a) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia\frac{y}{\lambda}} f(y) \frac{1}{|\lambda|} dy \\
&= \frac{1}{|\lambda|} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia\frac{y}{\lambda}} f(y) dy \\
&= \frac{1}{|\lambda|} \hat{f}\left(\frac{a}{\lambda}\right)
\end{aligned}$$

Remark A little interesting thing that we can see here from Analysis II is the absolute value here in the λ. The conclusion we have here is that in any case we have an absolute value. The same as when we were doing a change of variable in $\mathbb{R}^n$ where we took the absolute value of the determinant of the Jacobian.

If we were to visualize what scaling with a λ this would give our signal something like this:

Derivatives Let $g(x) = f'(x)$ then

$$\hat{g}(a) = ia\hat{f}(a)$$

More generally

$$\mathcal{F}\left[f^{(n)}\right](a) = (ia)^n \mathcal{F}[f](a)$$

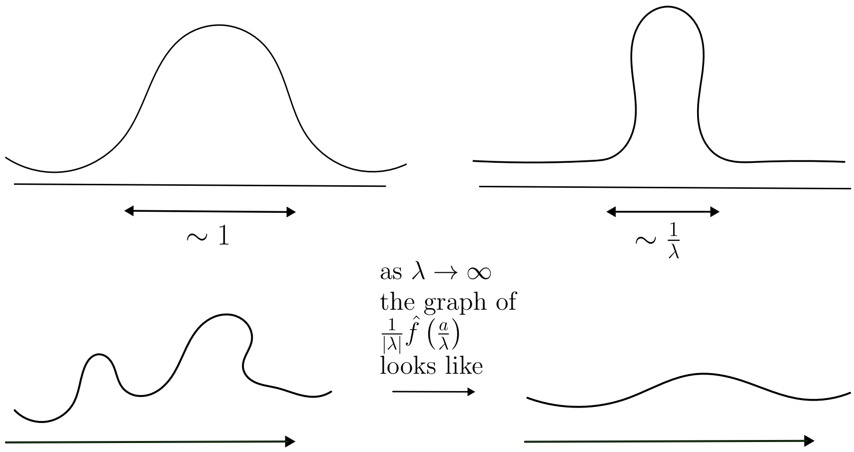


Figure 5.1: Figures on the difference between $f(x)$ and $f(\lambda x)$ and the Fourier transform of each function resp.

So the Fourier transform converts differential equations with constant coefficients into algebraic equations on a frequency domain. And this is really one of the big insight, the big feature of this transform.

Convolution

Definition 27. Given $f, g : \mathbb{R} \to \mathbb{C}$, the convolution $f * g : \mathbb{R} \to \mathbb{C}$, whenever is well-defined is:

$$f * g(x) = \int_{-\infty}^{\infty} f(y)g(x-y)dy$$

Convolution is sometime useful in order to '*smooth out a function*', the German equivalent has maybe a better illustration about smoothing by blablla. But first we shall see some properties of the convolution:

- Symmetry: $f * g = g * f$
- Associativity: $f * (g * h) = (f * g) * h$
- Distributivity $f * (g + h) = f * g + f * h$
- $(f * g)' = f' * g = f * g'$ whenever f or/and g are differentiable

The last item here is very important, it implies that $f * g$ is at least as smooth as the smoothest function among f and g. Interpreted as a *local average* of f with weight g.

Example

Let

$$g(x) = \begin{cases} \frac{1}{2\varepsilon} & \text{if } x \in [-\varepsilon, \varepsilon] \\ 0 \text{otherwise} \end{cases}$$

Now if we want to compute the convolution of a function f:

$$\begin{aligned} f * g(x) &= \int_{-\infty}^{\infty} f(y)g(x-y)dy \\ &= \frac{1}{2\varepsilon}\int_{x-\varepsilon}^{x+\varepsilon} f(y)dy \end{aligned}$$

So $f * g$ is really the local average of f. Note as $\varepsilon \to 0$ we have $f * g(x) \to f(x)$ when f is continuous at x.

Fourier transform as convolution

Let $f, g : \mathbb{R} \to \mathbb{C}$ then:

$$\mathcal{F}[f * g] = \sqrt{2\pi}\mathcal{F}[f]\mathcal{F}[g]$$

So the Fourier transform turns convolutions into products (and vis versa). In applications, one can often express $\mathcal{F}[f]$ as a product of Fourier transforms, which allows us to recover f by convolution.

Gaussian function

Now let us see the last property. If $f(x) = e^{\frac{-x^2}{2}}$ then the Fourier transform is:

$$\hat{f}(a) = e^{-\frac{a^2}{2}}$$

Will be shown in the exercises, but for the sake of completeness

Plancherel's Theorem

Theorem 14.

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} \left|\hat{f}(a)\right|^2 da$$

Okay this is nice but Fourier transform is about real function (as we have done them in Analysis III), how does complex analysis enter in the computation of Fourier transform?

Example

Let $f(x) = \frac{1}{1+x^2}$ then

$$\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} \frac{1}{1+x^2} dx$$

we can compute this using then the residue theorem!!!!! Let $a \le 0$. And set

$$f(z) = \frac{e^{-iaz}}{1+z^2} = \frac{e^{-iaz}}{(z-i)(z+i)}$$

Then in this situation, we will do the same as what we did in the previous section. Decompose the integral with a big part of the real axis, from $-R$ to R and close using the half circle. In our case we will choose the upper half circle, the reason is that the goal of this is that when $R \to \infty$ the contribution this half circle has to vanish.

Since $a \le 0$, when $z \in \{|z| = R\} \cap \{\text{Im } z > 0\}$ The function f decays exponentially as $R \to \infty$.

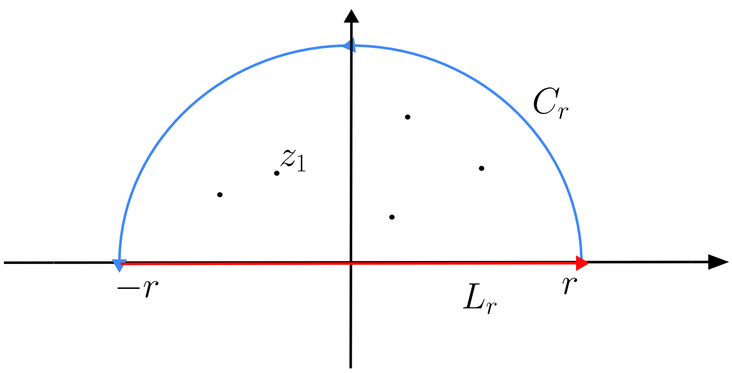


Figure 5.2: I am using a previous figure so don't mind the z in this

From previous calculations in class,

$$\int_{-\infty}^{\infty} f(x)dx = 2\pi i \text{Res}_i(f)$$

And we can check quite easily that i is a pole of order 1, we can therefore compute the residue using the limit:

$$= 2\pi i \lim_{z \to i}((z-i)f(z))$$

$$= 2\pi i \frac{e^a}{2i} = \pi e^{-|a|}$$

The reason of $-|a|$ is to suggest that the exponent is negative.

So for $a \leq 0$ and without forgetting the constant factor $\frac{1}{\sqrt{2\pi}}$, we have that the transform is given as

$$\mathcal{F}\left[\frac{1}{1+x^2}\right](a) = \sqrt{\frac{\pi}{2}}e^{-|a|}$$

Using similar calculation for $a > 0$ by considering the semi-circle in the lower half plane we will get the same result.

> *Remark* The principle is that in order to close the circle we choose the half circle depending on whether or not $a > 0$.

Chapter 6

Laplace Transform

The Fourier transform is made in order to computed for signals on $\mathbb{R}$. On the other hand the Laplace transform is made in order to compute signals defined on $\mathbb{R}_+$. Both transform differential equation into algebraic equation. The reason on why Laplace transform even is a thing (because we could think that every function defined for Laplace transform can be translated on $\mathbb{R}$ (e.g., setting $f(x) = 0 \forall x < 0$))? The reason for this is **convergence**.

Definition 28. Let $f : \mathbb{R}_+ \to \mathbb{C}$. When well-defined, the **Laplace transform** of f at $z \in \mathbb{C}$ is given as:

$$\mathcal{L}[f](z) := \int_0^\infty f(t)e^{-zt}dt$$

Sometimes Laplace transform of functions are written by capital letters

6.1 Convergence

Of course the above integral is well-defined if $\int_{-\infty}^{\infty} |f(t)||e^{-zt}|dt < \infty$.

Assume that there is a $\gamma_0 \in \mathbb{R}$ s.t.

$$\int_0^\infty |f(t)|e^{-\gamma_0 t}dt < \infty$$

What we are claiming is that this condition is sufficient in order to the Laplace transform to be defined. Naively this means that $|f(t)| < e^{\gamma_0 t}$ as $t \to \infty$

Using the triangle inequality:

$$\begin{aligned}
|\mathcal{L}[f](z)| &\leq \int_0^\infty |f(t)|\left|e^{-zt}\right|dt \\
&= \int_0^\infty |f(t)|\left|e^{-\text{Re } zt}\right| \underbrace{\left|e^{-\text{Im } zti}\right|}_{=1} dt \\
&= \int_0^\infty |f(t)|e^{-\text{Re } zt}dt \\
&\leq \int_0^\infty |f(t)|e^{-\gamma_0 t}dt < \infty
\end{aligned}$$

This mean that if such a γ_0 exists, $\mathcal{L}[f](z)$ is defined at least for $z \in \{z \in \mathbb{C} : \text{Re } z > \gamma_0\}$. Such a γ_0 has a name:

Definition 29. abscissa of convergence A $\gamma_0 \in \mathbb{R}$ such that $\mathcal{L}[f](z)$ is well defined at least on $\{\text{Re} > \gamma_0\}$ is called abscissa of convergence.

Remark 20. Remark for me about the triangle inequality, the triangle inequality is given as

$$|a+b| \leq |a| + |b|$$

By extending it to finite sum:

$$\left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|$$

As an integral is essentially a limit of sums, the same idea carries over:

$$\left|\int_a^b g(t)dt\right| \leq \int_a^b |g(t)|dt$$

Taking the modulus inside can only make things larger or equal, never smaller.

Example If $f(t) = 0$ for $t \geq M$ and f is piecewise continuous then the Laplace transform can be computed as:

$$\int_0^\infty |f(t)|e^{-\gamma_0 t}dt = \int_0^M |f(t)|e^{-\gamma_0 t}dt < \infty \qquad \forall \gamma_0 \in \mathbb{R}$$

In this case, $\mathcal{L}[f](z)$ is defined for all $z \in \mathbb{C}$

Let us have another example. Let γ_0 be an abscissa of convergence. Then $\mathcal{L}[f]$ is holomorphic on $\{\text{Re } > \gamma_0\}$. Set $F(z) = \mathcal{L}[f](z)$. Then:

$$\begin{aligned} \frac{d}{dz}F(z) &= \frac{d}{dz}\int_0^\infty f(t)e^{-zt}dt \\ &= \int_0^\infty f(t)(-t)e^{-zt} = -\mathcal{L}[tf(t)](z) \end{aligned}$$

In short,

$$\mathcal{L}[f(t)]' = -\mathcal{L}[tf(t)]$$

Remark Here we have implicitly used that if $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for $\mathcal{L}[f(t)]$ so it is for $\mathcal{L}[tf(t)]$.

Example II Let $f(z) = 1$. Then for every $\gamma_0 > 0$

$$\int_0^\infty |f(t)|e^{-\gamma_0 t}dt = \int_0^\infty e^{-\gamma_0 t}dt = \frac{1}{\gamma_0} < \infty$$

So any $\gamma_0 > 0$ is an abscissa of convergence, hence $\mathcal{L}[f](z)$ is defined for $z \in \{\text{Re } > 0\}$. Moreover:

$$\mathcal{L}[f](z) = \int_0^\infty e^{-zt}dt \underbrace{=}_{z\neq 0} \left[\frac{e^{-zt}}{-z}\right]_0^\infty$$

Note

$$\lim_{t\to\infty} e^{-zt} = 0$$

If Re $z > 0$ since:

$$\left|e^{-zt}\right| = \underbrace{\left|e^{-\text{Re } zt}\right|}_{\to 0}\underbrace{\left|e^{-\text{Im } zti}\right|}_{=1}$$

As $t \to \infty$

Therefore, $\mathcal{L}[1](z) = \frac{1}{z}$

Example Let us have $f : \mathbb{R}_+ \to \mathbb{C}, f(t) = t$. By the above derivative formula where Re $z > 0$

$$\frac{d}{dz}\underbrace{\mathcal{L}[i](z)}_{=\frac{1}{z}} = -\mathcal{L}[t](z) = -\mathcal{L}[f](z)$$

And thus

$$\mathcal{L}[t](z) = \frac{1}{z^2}$$

Iterating this procedure gives the following theorem

Theorem 15. For all $n \in \mathbb{N}$ and whenever Re $z > 0$

$$\mathcal{L}[t^n](z) = \frac{n!}{z^{n+1}}$$

6.2 Properties of the Laplace transform

Linearity Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be s.t. $\mathcal{L}[f], \mathcal{L}[g]$ are defined where Re $z > \gamma_0$ for some $\gamma_0 \in \mathbb{R}$. Then $\forall \lambda, \mu \in \mathbb{C}$ we have

$$\mathcal{L}[\lambda f + \mu g]$$

is defined when Re $z > \gamma_0$ and

$$\mathcal{L}[\lambda f + \mu g](z) = \lambda\mathcal{L}[f](z) + \mu\mathcal{L}[g](z)$$

Nomenclature Before diving into how to compute Laplace transform of derivative and/or Laplace transform. Let's us introduce a bit of notation:

Definition 30. We say $f : \mathbb{R}_+ \to \mathbb{C}$ is of class C^k, $k \in \mathbb{N}$, if f is k-times differentiable and $f, f', f'', \dots, f^{(k)}$ are continuous. (suffices to requires: $f^{(k)}$ continuous)

Theorem 16. Let $f : \mathbb{R}_+ \to \mathbb{C}$ be of class C^1. Assume $\gamma_0 \in \mathbb{R}$ abscissa of convergence for f and f'. Then for Re $z > \gamma_0$,

$$\mathcal{L}[f'](z) = z\mathcal{L}[f](z) - f(0)$$

Remark 21. Similarly to the Fourier transform, the Laplace transform converts derivatives into multiplication with z (hence differential equations into algebraic equations). A **crucial difference** is that the Laplace transform also picks up the '*initial value*' at 0.

Let us outline the proof of this theorem.

Proof. *Recall the definition of the Laplace transform:*

$$\mathcal{L}[f'](z) = \int_0^\infty f'(t)e^{-tz}dt$$

A derivative times an exponential? This looks a lot like an integration by part... However we have to be careful to treat the 'boundary term'at $\pm\infty$ appropriately. Since γ_0 is an abscissa of convergence for f' and $Re\ z > \gamma_0$, the integral $\int_0^\infty f'(t)e^{-tz}dt$ converges absolutely.

So for any sequence $T_n \to \infty$ as $n \to \infty$, Let us compute the Laplace transform

$$\mathcal{L}[f'](z) = \lim_{n\to\infty} \int_0^{T_n} f'(t)e^{-tz}dt$$

Using this trick we can now integrate by part

$$\begin{aligned}
&= \lim_{n\to\infty} \left[f(t)e^{-tz}\right]_0^{T_n} - \int_0^{T_n} f(t)(-z)e^{-tz}dt \\
&= \lim_{n\to\infty} f(T_n)e^{-T_n z} - f(0)\underbrace{e^{-0z}}_{=1} + z\int_0^\infty f(t)e^{-tz}dt \\
&= \lim_{n\to\infty} f(T_n)e^{-T_n z} - f(0) + z\mathcal{L}[f](z)
\end{aligned}$$

Now we should choose a sequence T_n such that the latter limit is 0. This is possible because, since γ_0 is an abscissa of convergence for f and $Re\ z > \gamma_0$

$$\int_0^\infty |f(t)|\left|e^{-zt}\right|dt \le \int_0^\infty |f(t)|e^{-Re\ zt}dt < \infty$$

Thus there must exists a sequence $T_n \to \infty$ as $n \to \infty$ s.t.

$$|f(T_n)|\left|e^{-zT_n}\right| \to 0 \text{ as } n \to \infty$$

The argument for that is that otherwise $|f(t)||e^{-zt}|$ would stay away from 0 as $t \to \infty$ but such function cannot have finite integral over the positive line.
When T_n is chosen that way, the computation above gives us

$$\mathcal{L}[f'](z) = -f(0) + z\mathcal{L}[f](z)$$

□

As a consequence of iterating this formula: if $f : \mathbb{R}_+ \to \mathbb{C}$ is of class C^k for $k \in \mathbb{N}$ and $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for all derivatives of f namely $f, f', f'', \ldots, f^{(k)}$ then for Re $z > \gamma_0$,

$$\begin{aligned}
\mathcal{L}[f'](z) &= z\mathcal{L}[f](z) - f(0) \\
\mathcal{L}[f''](z) &= z\mathcal{L}[f'](z) - f'(0) \\
&= z[z\mathcal{L}[f] - f(0)] - f'(0)
\end{aligned}$$

By iterating this formula for k, this gives us

$$\mathcal{L}\left[f^{(k)}\right](z) = z^k\mathcal{L}[f](z) - \sum_{j=0}^{k-1} z^{k-1-j}f^{(j)}(0)$$

Example Consider $f(t) = t^2, f'(t) = 2t, f''(t) = 2$. As computed above, the Laplace transform are defined when Re $z > 0$
Which confirms our previous computations. Note carefully we used $f(0) = f'(0) = 0$

Theorem 17. Let $f : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous and $\gamma_0 \ge 0$ an abscissa of convergence for f. Define $g : \mathbb{R}_+ \to \mathbb{C}$ by $g(t) = \int_0^t f(s)ds$. Then when Re $z > \gamma_0$,

$$\mathcal{L}[g](z) = \frac{1}{z}\mathcal{L}[f](z)$$

$$\begin{tikzcd}
f & \mathcal{L}[f](z) = \frac{2}{z^3} \arrow[d, bend right, "\cdot z"'] \\
f' & \mathcal{L}[f'](z) = \frac{2}{z^2} \arrow[u, bend right, "\cdot \frac{1}{z}"'] \arrow[d, bend right, "\cdot z"'] \\
f'' & \mathcal{L}[f''](z) = \frac{2}{z} \arrow[u, bend right, "\cdot \frac{1}{z}"']
\end{tikzcd}$$

Figure 6.1: Laplace transform of polynomials

Remark 22. In general the abscissa of convergence for g cannot be negative. (even if it is for f).

Proof. *Note $g' = f$ and $g(0) = 0$. By the previous theorem,*

$$\mathcal{L}[f](z) = \mathcal{L}[g'](z) = z\mathcal{L}[g](z) - \underbrace{g(0)}_{=0}$$

Therefore we have

$$\mathcal{L}[g](z) = \frac{1}{z}\mathcal{L}[f](z)$$

$\square$

(We will not show why $\mathcal{L}[g](z)$ is indeed defined for Re $z \geq \gamma_0$ when $\gamma_0 \geq 0$)

Theorem 18. Let $a > 0$ and $b \in \mathbb{R}$. Let $f : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous with an abscissa of convergence $\gamma_0 \in \mathbb{R}$. Define $g : \mathbb{R}_+ \to \mathbb{C}$ by $g(t) = e^{-bt}f(at)$. Then

$$\mathcal{L}[g](z) = \frac{1}{a}\mathcal{L}[f]\left(\frac{z+b}{a}\right)$$

This hold in essence when the right hand side exists, and this is the case when Re $\frac{z+b}{a} > \gamma_0$ (which is equivalent to Re $z > a\gamma_0 - b$ because $a > 0$ and $b \in \mathbb{R}$)

In fact: $a\gamma_0 - b$ is an abscissa of convergence for g

Proof. *By the definition we have that*

$$\begin{aligned}
\mathcal{L}[g](z) &= \int_0^\infty e^{-bt}f(at)e^{-zt}dt \\
&= \int_0^\infty f(at)e^{-(z+b)t}dt
\end{aligned}$$

So here in order to have the same transformation as the one for f we have to get rid of the a in the argument, we can do so using substitution. Let $s = at$ and $ds = adt$. As we know that $a > 0$ we can rewrite our integral as

$$\begin{aligned}
&= \int_0^\infty f(s)e^{(-z+b)\frac{s}{a}}\frac{ds}{a} \\
&= \frac{1}{a}\int_0^\infty f(s)e^{-\frac{z+b}{a}s}ds = \frac{1}{a}\mathcal{L}[f]\left(\frac{z+b}{a}\right)
\end{aligned}$$

$\square$

Example Let us have $g(t) = e^{ct} = e^{ct} \cdot 1$. Using the fact that the Laplace transform of 1 is $\frac{1}{z}$ we can use the theorem stated above in order to find the Laplace transform of g. Here $a = 1, b = -c$. By the theorem, where $\text{Re } z > -b = c$ we have

$$\mathcal{L}[g](z) = \mathcal{L}[1](z - c) = \frac{1}{z - c}$$

Remark 23. The fact that c is an abscissa of convergence for g can be shown by hand, without relying on the above theorem.

Example Find $f : \mathbb{R}_+ \to \mathbb{C}$ such that $\mathcal{L}[f](z) = \frac{1}{(z-a)(z-b)}$ where $a, b \in \mathbb{R}$. To do so we will divide our transform into sum of fraction as in the previous example. Then using the linearity of the Laplace transformation we will be able to find our function.

$$\begin{aligned}
\mathcal{L}[f](z) &= \frac{1}{b-a}\left[\frac{1}{z-b} - \frac{1}{z-a}\right] \\
&= \frac{1}{b-a}\left[\mathcal{L}\left[e^{bt}\right](z) - \mathcal{L}\left[e^{at}\right](z)\right] \\
&= \mathcal{L}\left[\frac{e^{bt} - e^{at}}{b-a}\right](z)
\end{aligned}$$

This is defined when the two Laplace transforms in the middle are both defined, when $\text{Re } z > \max\{a, b\}$.

Convolution Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ functions that are piecewise continuous. By extending those functions by 0 on $(-\infty, 0)$. We can again defined the convolution as we did in previous section

$$f * g(t) = \int_{-\infty}^{\infty} f(t-s)g(s)ds = \begin{cases} \int_0^t f(t-s)g(s)ds & t > 0 \\ 0 & \text{otherwise} \end{cases}$$

The reason for the case is that we can cut out bornes on our integral. Notice that $g(s) = 0$ when $s < 0$ therefore all the interval $(-\infty, 0)$ doesn't have any effect on the result of our integral (as $f(t-s)g(s) = 0$ there). Also, when $s > t$, the argument of f becomes negative which also makes the value integrated 0. Therefore only when $s \in (0, t)$ we have a non trivial contribution to our integral.

In particular $f * g$ can be restricted back to $\mathbb{R}_+$ since its values on $(-\infty, 0)$ are trivial. That way we can regard $f * g : \mathbb{R}_+ \to \mathbb{C}$ and compute its Laplace transform.

Theorem 19. Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous with common abscissa of convergence $\gamma_0 \in \mathbb{R}$. Then γ_0 is an abscissa of convergence for $f * g$ and when $\text{Re } z > \gamma_0$

$$\mathcal{L}[f * g](z) = \mathcal{L}[f](z)\mathcal{L}[g](z)$$

6.3 Applications of Laplace analysis to the Cauchy problem (initial value problem)

With the above formulas we can '*invert*' the Laplace transform of many functions. That is the say, if we have a function that is the Laplace function of something, there is a lot of case were can we find what is **something** is. Note that we will also see the formal definition of the Laplace transform.

6.3.1 Ordinary Differential Equations (ODE) with constant coefficients

First order initial value problem The first type of equations we will talk about is first order initial value problem. Formally we have:

Let $a, y_0 \in \mathbb{R}$ and $f : \mathbb{R}_+ \to \mathbb{R}$ be piecewise continuous. The task is to find a solution $y : \mathbb{R}_+ \to \mathbb{R}$ solving

$$\begin{cases} y'(t) + ay(t) = f(t) \quad \forall t > 0 \\ y(0) = y_0 \end{cases}$$

We will address two question

- **existence**: whether there is a solution or not (this will be done using Laplace analysis).
- **uniqueness**: there is a unique solution

Remark In general, we will have linear initial value problems (i.e., no non linear dependency on the sought solution y such as $y(t)^2, y(t)^3$, etc.). Also, if the number of initial conditions is equal to the order of the equation then the solution is unique. (e.g., $y(0) = y_0$ above)

Proof. *Now we will prove the uniqueness of the equation written above. To do so let us have two solutions $y_1, y_2 : \mathbb{R}_+ \to \mathbb{R}$, the goal is to show that they coincide. We will do this by showing that $w(t) = 0 \forall t \in \mathbb{R}_+$, where $w(t) = y_1(t) - y_2(t)$.*

First we know that both y_1 and y_2 satisfy the equation, using that we will try to derive a differential equation for w. In this case this is easy as because w solve this differential equation (as this is linear combinations).

$$\begin{cases} w'(t) + aw(t) &= 0 \\ w(0) &= 0 \end{cases}$$

*It is pretty easy to see that $w(t) = 0$ is a solution, but we also need to prove that it is the **only** solution. To do so we will multiply w with e^{at}. Then the previous equation gives*

$$\begin{aligned} \left(e^{at} w(t)\right)' &= ae^{at} w(t) + e^{at} w'(t) \\ &= ae^{at} w(t) - ae^{at} w(t) = 0 \end{aligned}$$

Hence $e^{at} w(t)$ is constant in $t \in \mathbb{R}_+$. And as $e^{at} w(t) = e^{a0} w(0) = 0$, this shows that $w(t) = 0$ for all $t \in \mathbb{R}_+$. ☐

Now let us proof **Existence**

Proof. *To do so we will use the Laplace transform of the differential equation*

$$y'(t) + ay(t) = f(t)$$

Let us apply Laplace (we will be deliberate on the existence of the Laplace transform here)

$$\mathcal{L}[y'](z) + a\mathcal{L}[y](z) = \mathcal{L}[f](z)$$

Let us now set $Y = \mathcal{L}[y]$ and $F = \mathcal{L}[f]$ and using the formula for the Laplace transform of derivative we have:

$$zY(z) - \underbrace{y(0)}_{=y_0} + aY(z) = F(z)$$

By solving this we get

$$Y(z) = \frac{F(z)}{z+a} + \frac{y_0}{z+a}$$

We use the punchline: 'derivatives transform derivatives equations to algebraic equations'

The question we need to answer is, does the right hand side of the above expression is a Laplace transform of something? To do so we can recall that:

$$\frac{1}{z+a} = \mathcal{L}\left[e^{-at}\right](z)$$

This implies that we get

$$Y(z) = \mathcal{L}\left[e^{-at}\right](z)\mathcal{L}[f](z) + y_0\mathcal{L}\left[e^{-at}\right](z)$$

We can see here that the first part of the right hand side is the Laplace transform of the convolution of e^{-at} with $f(t)$! This would 'implies' that

$$y(t) = \int_0^t e^{-a(t-s)} f(s)ds + y_0 e^{-at}$$

Indeed one can verify that this is a solution of the differential equation. $\square$

Second order initial value problem

Now let us go into the second order. To do so we will define it as:

Definition 31. Let $w \in \mathbb{R}_+, a > 0$, and $f : \mathbb{R}_+ \to \mathbb{R}$ as above

$$\begin{cases} y''(t) + 2ay'(t) + w^2 y(t) &= f(t) \qquad \forall t > 0 \\ y(0) &= y_0 \\ y'(0) &= v_0 \end{cases}$$

Remark This is the equation satisfied by the harmonic oscillator, f is representing a forcing parameter

As in the first order case we also have uniqueness

Uniqueness **Proof.** *Let $y_1, y_2 : \mathbb{R}_+ \to \mathbb{R}$ be two solutions of the ODE. Let us set $w(t) = y_1(t) - y_2(t)$. As above we want to show that $w(t) = 0 \quad \forall t \in \mathbb{R}_+$ by using* ***energy method****.*

To do so, we will subtract in the ODE y_2 from y_1 this gives us

$$\begin{cases} w''(t) + 2aw'(t) + \omega^2 w(t) &= 0 \\ w(0) &= 0 \\ w'(0) &= 0 \end{cases}$$

Let us now magically multiply the equation by w':

$$\underbrace{u''u'}_{=\frac{1}{2}\left((u')^2\right)'} + 2a(u')^2 + \omega^2 \underbrace{u'u}_{=\frac{1}{2}(u^2)'} = 0$$

$$\implies \left(\frac{1}{2}(w')^2\right)' + 2a(w')^2 + \left(\frac{1}{2}\omega^2 w^2\right)'$$

And as we can see, the left part here is the kinetic energy and the right part is the potential energy, this is why this is called the energy method.
Then this implied that:

$$\left(\frac{1}{2}(u')^2 + \frac{1}{2}\omega^2 u^2\right)' = -2a(u')^2 \leq 0$$

This means that the energy $E(t) = \frac{1}{2}u'(t)^2 \frac{1}{2}\omega^2 u(t)^2$ *is* ***nonincreasing*** *(effect of damping). Since the energy of* $E(0) = 0$ *(because* $u(0) = 0$ *and* $u'(0) = 0$*) and* $E(t) \geq 0$ *(since it is a sum of squares) So we are looking at a function that is nonincreasing and that starts at* 0 ***but*** *is positive. This implies that*

$$u(t) = 0 \qquad \forall t \in \mathbb{R}_+$$

Since $0 \leq \frac{1}{2}\omega^2 u^2 \leq \frac{1}{2}\omega^2 u^2 + \frac{1}{2}(u')^2 = E = 0$ *Which implies that* $u^2 = 0$ □

Existence

Proof. *We will again apply the Laplace transform to the equation*

$$y''(t) + 2ay'(t) + \omega^2 y(t) = f(t)$$

We get

$$z^2 Y(z) - z\underbrace{y(0)}_{=y_0} - \underbrace{y'(0)}_{v_0} + 2a[zY(z) - y_0] + \omega^2 Y(z) = F(z)$$

This is just an algebraic equation for Y *therefore by rearranging it, this gives:*

$$\left(z^2 + 2az + \omega^2\right)Y(z) = F(z) + (z+1)y_0 + ay_0 + v_0$$

By completing the square:

$$\begin{aligned} z^2 + 2az + \omega^2 &= (z+a)^2 + \omega^2 - a^2 \\ \implies Y(z) = \frac{1}{(z+a)^2 + \omega^2 - a^2}F(z) &+ \frac{z+a}{(z+a)^2 + \omega^2 - a^2}y_0 + \\ &\frac{1}{(z+a)^2 + \omega^2 - a^2}(ay_0 + v_0) \end{aligned}$$

we now consider three different cases depending on the sign of $\omega^2 - a^2$

1. $a^2 = \omega^2$ *(critically damped). In this case this simplifies as*

$$\frac{1}{(z+a)^2 + \omega^2 - a^2} = \frac{1}{(z+a)^2}$$
$$\frac{z+a}{(z+a)^2 + \omega^2 - a^2} = \frac{1}{z+a}$$

Recall that $\mathcal{L}[e^{-at}](z) = \frac{1}{z+a}$ *and also that*

$$\begin{aligned} \frac{1}{(z+a)^2} = -\frac{d}{dz}\frac{1}{z+a} &= -\frac{d}{dz}\mathcal{L}\left[e^{-at}\right](z) \\ &= \mathcal{L}\left[te^{-at}\right](z) \end{aligned}$$

This implies that

$$\begin{aligned} Y(z) = \mathcal{L}\left[te^{-at}\right](z)\mathcal{L}[f](z) + \mathcal{L}\left[e^{-at}\right](z)y_0 \\ + \mathcal{L}\left[te^{-at}\right](z)(ay_0 + v_0) \end{aligned}$$

By formally inverting this will gives us (again using convolution)

$$y(t) = \int_0^t (t-s)e^{-a(t-s)}f(s)ds + e^{-at}y_0 + te^{-at}(ay_0 + v_0)$$

2. $\omega^2 > a^2$ *(underdumped). This time we start from*

$$\mathcal{L}[\cos(\beta t)](z) = \frac{z}{z^2+\beta^2}$$
$$\mathcal{L}[\sin(\beta t)](z) = \frac{\beta}{z^2+\beta^2}$$

Define $\beta = \sqrt{\omega^2 - a^2}$ *(this is totally legal as we have the hypothesis above). Again let us rewrite the equations in order to make appear the reciprocals*

$$\begin{aligned}\frac{1}{(z+a)^2+\beta^2} &= \frac{1}{\beta}\frac{\beta}{(z+a)^2+\beta^2}\\ &= \frac{1}{\beta}\mathcal{L}\big[e^{-at}\sin(\beta t)\big](z)\end{aligned}$$

In order to show this we use the frequency shift property that we have state in previous sections. For the other parts of the equations

$$\frac{z+a}{(z+a)^2+\beta^2} = \mathcal{L}\big[e^{-at}\cos(\beta t)\big](z)$$

This gives us

$$\begin{aligned}Y(t) &= \mathcal{L}\left[\frac{1}{\sqrt{\omega^2-a^2}}e^{-at}\sin\Big(\sqrt{\omega^2-a^2}t\Big)\right](z)\mathcal{L}[f](z)\\ &+ \mathcal{L}\Big[e^{-at}\cos\Big(\sqrt{\omega^2-a^2}t\Big)\Big](z)y_0\\ &+ \mathcal{L}\left[\frac{1}{\sqrt{\omega^2-a^2}}e^{-at}\sin\Big(\sqrt{\omega^2-a^2}t\Big)\right](z)(ay_0+v_0)\end{aligned}$$

Therefore

$$\begin{aligned}y(t) &= \int_0^t \frac{1}{\sqrt{\omega^2-a^2}}e^{-a(t-s)}\sin\Big(\sqrt{w^2-a^2}(t-s)\Big)f(s)ds\\ &+ y_0e^{-at}\cos\Big(\sqrt{\omega^2-a^2}t\Big)\\ &+ (ay_0+v_0)\frac{1}{\sqrt{\omega^2-a^2}}e^{-at}\sin\Big(\sqrt{\omega^2-a^2}t\Big)\end{aligned}$$

3. $\omega^2 < a^2$ *(overdumped) Let us recall the transform*

$$\mathcal{L}[\cosh(\beta t)](z) = \frac{z}{z^2-\beta^2}$$
$$\mathcal{L}[\sinh(\beta t)](z) = \frac{\beta}{z^2-\beta^2}$$

This can be shown using the fact that $\cosh(x) = \cos(ix)$ *and* $\sinh(x) = -i\sin(ix)$

Using the definition $\beta = \sqrt{a^2-\omega^2}$, *then*

$$\begin{aligned}\frac{1}{(z+a)^2+\omega^2-a^2} &= \frac{1}{\beta}\frac{\beta}{(z+a)^2-\beta^2}\\ &= \frac{1}{\beta}\mathcal{L}\big[e^{-at}\sinh(\beta t)\big](z)\end{aligned}$$

Using the same technique we get

$$\frac{z+a}{(z+a)^2-\beta^2} = \mathcal{L}\left[e^{-at}\cosh(\beta t)\right](z)$$

By again inserting this in the Laplace transform we get

$$\begin{aligned} Y(z) &= \mathcal{L}\left[\frac{1}{\sqrt{a^2-\omega^2}}e^{-at}\sinh\left(\sqrt{a^2-\omega^2}t\right)\right](z)\mathcal{L}[f](z) \\ &\quad + \mathcal{L}\left[e^{-at}\cosh\left(\sqrt{a^2-\omega^2}t\right)\right](z)y_0 \\ &\quad + \mathcal{L}\left[\frac{1}{\sqrt{a^2-\omega^2}}e^{-at}\sinh\left(\sqrt{a^2-\omega^2}t\right)\right](z)(ay_0+v_0) \end{aligned}$$

Therefore

$$\begin{aligned} y(t) &= \int_0^t \frac{1}{\sqrt{a^2-\omega^2}}e^{-a(t-s)}\sinh\left(\sqrt{a^2-\omega^2}(t-s)\right)f(s)ds \\ &\quad + y_0e^{-at}\cosh\left(\sqrt{a^2-\omega^2}t\right) \\ &\quad + (ay_0+v_0)\frac{1}{\sqrt{a^2-\omega^2}}e^{-at}\sinh\left(\sqrt{a^2-\omega^2}t\right) \end{aligned}$$

□

The reason on why we did this is to show **how** to solve those type of equations, using Laplace transform analysis and properties about them.

6.3.2 Inverse Laplace Transform

This is the same principle as how we transformed the Fourier transform, the question we are trying to answer is: '*Given $\mathcal{L}[y]$ can we determine y in general?*' To do so we will need to recall some theory: if $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for f then $\mathcal{L}[f]$ is holomorphic (hence, without singularities) On $\{\text{Re } z > \gamma_0\}$. In fact the '*opposite*' is also true: if $\mathcal{L}[f]$ is holomorphic on $\{\text{Re } z > \gamma_0\}$ and '*doesn't grow*' as $|z| \to \infty$ then γ_0 is an abscissa of convergence for f.

Theorem 20. Assume $f : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous for some $\gamma \in \mathbb{R}, \mathcal{L}[f]$ is holomorphic on $\{\text{Re } z \geq \gamma\}$ and

$$\lim_{T\to\infty}\int_{-T}^{T} F(\gamma+is)e^{(\gamma+is)t}ds$$

exists for all $t \in \mathbb{R}_+$. (Both hold when there exists $\gamma_0 < \gamma$ abscissa of convergence). Then

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathcal{L}[f](\gamma+is)e^{(\gamma+is)t}ds$$

What this gives us is geometrically is

How to show this formula? (Out-book)

Let us extend $f : \mathbb{R}_+ \to \mathbb{C}$ to a function defined on $\mathbb{R}$ by setting it to be 0 on $(-\infty, 0)$. Let us define

$$g(t) = f(t)e^{-\gamma t}$$

And let us take the Fourier transform of g

$$\begin{aligned} \hat{g}(a) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} g(t)e^{-iat}dt \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{-(\gamma+ia)t}dt \\ &= \frac{1}{\sqrt{2\pi}}\int_{0}^{\infty} f(t)e^{-(\gamma+ia)t}dt = \mathcal{L}[f](\gamma+ia) \end{aligned}$$

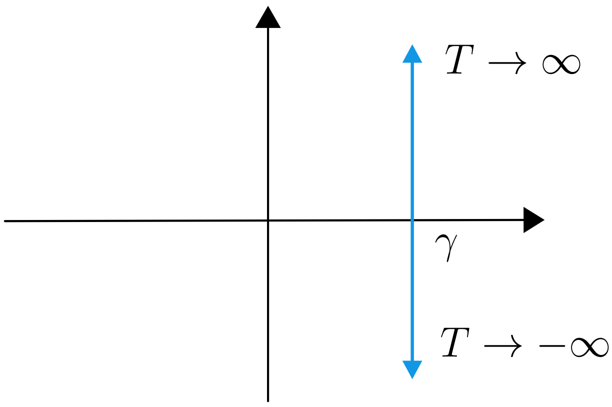


Figure 6.2: The blue pale thing here is the what we are integrating

Now let us take the inverse of the Fourier transform

$$\begin{aligned} g(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{g}(a) e^{iat} da \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\gamma + ia) e^{iat} da \\ \implies f(t) = e^{\gamma t} g(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\gamma + ia) e^{(\gamma + ia)t} dt \end{aligned}$$

Remark 24.
1. Lercle's theorem tells us that f is uniquely determined by $\mathcal{L}[f]$ through this formula
2. The condition $\lim_{T\to\infty} \int_{-T}^{T} F(\gamma + is) e^{(\gamma+is)t} ds < \infty$ is given when provided that

$$\int_{-\infty}^{\infty} |F(\gamma + is)| ds < \infty$$

Example Let us have the function

$$F(z) = \frac{1}{(z+1)(z+2)^2}$$

The goal is to find a $f : \mathbb{R}_+ \to \mathbb{C}$ such that $\mathcal{L}[f] = F$. To do so we have two ways.

Partial fraction decomposition Using partial fraction decomposition, we can simplify our function into a sum of easier function

$$\frac{1}{(z+1)(z+2)^2} = \underbrace{\frac{1}{z+1}}_{=\mathcal{L}[e^{-t}]} - \underbrace{\frac{1}{z+2}}_{=\mathcal{L}[e^{-2t}]} - \underbrace{\frac{1}{(z+2)^2}}_{=\mathcal{L}[te^{-2t}]}$$

And as the inverse of the Laplace transform of those function is known/ easy to find we get

$$f(t) = e^{-t} - e^{-2t} - te^{-2t}$$

And by **uniqueness** of the Laplace transform, this is enough to find f and it usually is the fastest way.

Inverse Laplace transform and residue theorem Using the inverse Laplace transform we know that f is given as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\gamma + is) e^{(\gamma+is)t} ds$$

Let $h(z) = F(z)e^{zt}$. The poles of our function h are $z = -1$ and $z = -2$. For simplicity, let us choose $\gamma = 0$. Now we have to check some properties before we can actually compute this integral, we have that

1. F is holomorphic on $\{\text{Re } z > -1\}$
2. $\int_{-\infty}^{\infty} |F(\gamma + is)| ds < \infty$

We have then the Inverse of the Laplace transform is given as

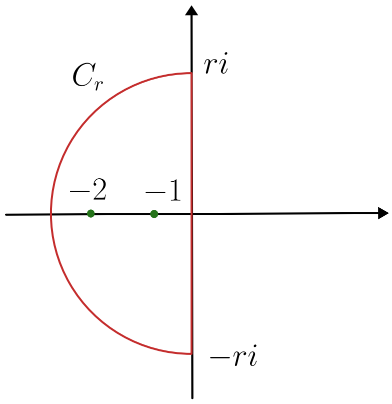
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(\gamma + is) ds = \frac{1}{2\pi i} \int_{-\infty i}^{\infty i} h(z) dz$$

Now we can use our usual trick of closing the curve in order to be able to use the residue theorem. As our residue are on the left side of the complex plane, let us use the half circle which is defined on the left

To do so we will define two curves

$$\begin{aligned} L_r : [-r, r] &\longmapsto \mathbb{C} \\ s &\longmapsto si \end{aligned}$$

$$\begin{aligned} C_r : \left[\frac{\pi}{2}, \frac{2\pi}{2}\right] &\longmapsto \mathbb{C} \\ \theta &\longmapsto re^{i\theta} \end{aligned}$$



This gives us the curve Γ we need as

$$\Gamma = L_r \cup C_r \quad \text{oriented counterclockwise}$$

When $r > 2$, Γ encloses all pole of h, namely $-1, -2$. Therefore now that we have done all the work (almost) to use the residue theorem, let us use it!

$$\int_\Gamma h(z) dz = 2\pi i [\text{Res}_{-1}(h) + \text{Res}_{-2}(h)]$$

Let us now computes the residue, the easiest one is $z = -1$ as it is a simple pole. We therefore only need to compute the limit given as

$$\text{Res}_{-1}(h) = \lim_{z \to -1} ((z+1)h(z)) = \lim_{z \to -1} \frac{e^{zt}}{(z+2)^2} = e^{-t}$$

For the second pole, $z = -2$ this pole is of order 2 which therefore implies that we need to take the derivative as

$$\begin{aligned} \text{Res}_{-2}(h) &= \lim_{z \to -2} \frac{d}{dz} (z+2)^2 h(z) \\ &= \lim_{z \to -2} \frac{d}{dz} \left(\frac{e^{zt}}{z+1} \right) = -e^{-2t}(t+1) \end{aligned}$$

Hence giving us the result

$$\int_\Gamma h(z)dz = 2\pi i\big[e^t - e^{-2t} - te^{-2t}\big]$$

What we need to show now is that the L_r part of the integral vanishes as $r \to \infty$, to do so we will use a bit of triangle inequality everywhere. The goal is to show that the integral

$$\left|\int_{C_r} h(z)dz\right| \to 0 \text{ as } r \to \infty$$

$$\begin{aligned}\left|\int_{C_r} h(z)dz\right| &= \left|\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} g\big(re^{i\theta}\big)ire^{i\theta}d\theta\right| \\ &\leq \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \big|h\big(re^{i\theta}\big)\big|rd\theta \\ &= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{\left|e^{re^{i\theta}}\right|}{|re^{i\theta}+1||re^{i\theta}+2|^2}rd\theta\end{aligned}$$

And here we can noting that the term $r \cdot \cdots$ dominates, the lower term of this integral can be approximates by $\sim r \cdot \sim r^2$. Now let us work with the upper term

$$\begin{aligned}\left|e^{re^{i\theta}}\right| &= \left|e^{r(\cos\theta + i\sin\theta)t}\right| \\ &= \left|e^{r\cos\theta t}\right|\Bigg|\underbrace{e^{ir\sin\theta t}}_{=1}\Bigg| = e^{r\cos\theta t} \leq 1\end{aligned}$$

Because $\cos\theta \leq 0$ for $\theta \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$. Hence, the integral $\left|\int_{C_r} h(z)dz\right|$ can be bounded from above by a term of order

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{rr^2}rd\theta \to 0 \text{ as } r \to \infty$$

Therefore, we have found our inverse given as

$$\frac{1}{2\pi i} \cdot 2\pi i\big[e^{-t} - e^{-2t} - te^{-2t}\big]$$

Remark 25. We chose the left half circle (for Re $z < \gamma$) because in that region $|e^{zt}| \leq e^{\gamma t}$. This guarantees the control on the upper term of h.

Chapter 7

Application to partial differential equations (PDES)

As for ODES we will apply the theory of Laplace and Fourier transform to solve initial boundary value problems for linear partial differential equations with constant coefficients, e.g. heat equation

$$\frac{\partial u}{\partial t}(x,t) - \frac{\partial^2 u}{\partial x^2}u(x,t) = f(x,t)$$

For $t > 0$ and $x \in (0, L)$. We stipulate (or give the initial condition)

$$\begin{aligned} u(x,0) &= u_0(x) \\ u(0,t) &= u(L,t) = 0 \quad \text{boundary condition} \end{aligned}$$

It describes the evolution of temperature u of a one dimensional rod of length L with a heat source f and fixed temperature at endpoints $0, L$.

Let us summarize before starting the general procedure that we will follow. There is three different '*tools*' that we can use:

- **Fourier series** if the space or time domain is bounded
- **Fourier transform** if the space or time domain is on $\mathbb{R}$
- **Laplace transform** if the space or time domain is half unbounded ($\mathbb{R}_+$)

7.0.1 Recall of Fourier series

What we want to decompose is a function $f : [0, L] \to \mathbb{R}$ into an infinite sum of oscillating functions. Those function are made using **basis function**, namely

$$\begin{aligned} &\sin\left(\frac{2\pi n}{L}x\right) \quad \text{where } n \geq 1 \\ &\cos\left(\frac{2\pi n}{L}x\right) \quad \text{where } n \geq 0 \end{aligned}$$

Orthogonality

Definition 32. Inner product The inner product between two function f, g is given as

$$< f, g >:= \int_0^L f(x)g(x)dx$$

$$< \sin\left(\frac{2\pi n}{L}x\right), \sin\left(\frac{2\pi n}{L}x\right) >= \begin{cases} \frac{L}{2} & \text{if } m = n \\ 0 & \text{otherwise} \end{cases}$$

The same foes for the product

$$< \cos\left(\frac{2\pi n}{L}x\right), \cos\left(\frac{2\pi n}{x}\right) >$$

Theorem 21.

$$< \sin\left(\frac{2\pi n}{L}x\right), \cos\left(\frac{2\pi n}{L}x\right) >= 0 \quad \forall n, m$$

As we can see those function creates an orthogonal basis which we can use in order to decompose other function.

The question that the Fourier series is answering is: Given $f : [0,1] \to \mathbb{R}$ be a piecewise continuous function, can we write it as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{2\pi n}{L}x\right) + b_n \sin\left(\frac{2\pi n}{L}x\right)\right]$$

Remark The same consideration apply when $f : \mathbb{R} \to \mathbb{R}$ is L-periodic, which can therefore be thought of as a function on $[0, L]$.

Coefficient of the serie In order to guess the form of the coefficients a_n and b_n, we will assume that f already has this form. Then we can formally compute the following using orthogonality

$$\int_0^L f(x)dx = \frac{L}{2}a_0$$

$$\int_0^L f(x)\cos\left(\frac{2\pi n}{L}x\right)dx = \frac{L}{2}a_n \quad \text{where } n \geq 1$$

$$\int_0^L f(x)\sin\left(\frac{2\pi n}{L}x\right)dx = \frac{L}{2}b_n \quad \text{where } n \geq 1$$

This suggests to define our coefficients as

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{2\pi n}{L}x\right)dx \quad \text{where } n \geq 0$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{2\pi n}{L}x\right)dx \quad \text{where } n \geq 1$$

Theorem 22. Define $f_N : [0, L] \to \mathbb{R}$ as

$$f_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N}\left[a_n \cos\left(\frac{2\pi n}{L}x\right) + b_n \sin\left(\frac{2\pi n}{L}x\right)\right]$$

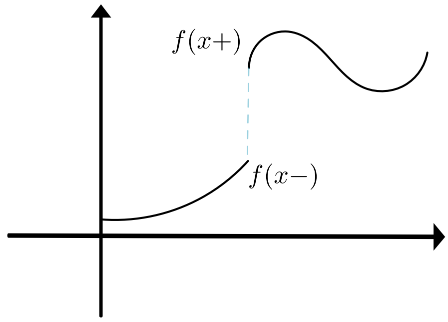
Where a_n and b_n are defined as above, where $f : [0, L] \to \mathbb{R}$ is piecewise continuous. Then

$$f_N(x) \to f(x) \text{ as } N \to \infty \quad \text{at every } x \in [0, L]$$

$$f_N(x) \to \frac{f(x+) + f(x-)}{2} \quad \text{when } x \text{ is a jump discontinuity of } f$$

$$\lim_{N\to\infty}\int_0^L |f_N(x) - f(x)|dx = 0$$

The definition of the $f(x+)$ and $f(x-)$ are designed in such that when we have a discontinuity, we are effectively taking the mean of those two points



Special cases

f is odd Assume f is odd, then

$$a_n = 0 \quad \forall n \geq 0$$

And therefore

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{2\pi n}{L} x\right)$$

Where

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L f(x) \sin\left(\frac{2\pi n}{L} x\right) dx \\ &= \frac{2}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} f(x) \sin\left(\frac{2\pi n}{L} x\right) dx \\ &= \frac{4}{L} \int_0^{\frac{L}{2}} f(x) \sin\left(\frac{2\pi n}{L} x\right) dx \end{aligned}$$

We are able to do so as f and sin are odd, their product is even.

f is even Assume f is even, then

$$b_n = 0 \quad \forall n \geq 1$$

In this case f is decomposed as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi n}{L} x\right)$$

Where

$$\begin{aligned} a_n &= \frac{2}{L} \int_0^L f(x) \cos\left(\frac{2\pi n}{L} x\right) dx \\ &= \frac{2}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} f(x) \cos\left(\frac{2\pi n}{L} x\right) dx \quad \text{by } L-\text{periodicity} \\ &= \frac{4}{L} \int_0^{\frac{L}{2}} f(x) \cos\left(\frac{2\pi n}{L} x\right) dx \end{aligned}$$

Remark 26.
- The function $a_n \cos\left(\frac{2\pi n}{L}x\right)$ and $b_n \sin\left(\frac{2\pi n}{L}x\right)$ are sometimes called **modes**
- The function $e_n(x) = \cos\left(\frac{2\pi n}{L}x\right)$ satisfies the Neunmann boundary condition

$$e_n'(0) = 0 \quad e_n'(L) = 0$$

- The function $h_n(x) = \sin\left(\frac{2\pi n}{L}x\right)$ satisfies **Dirichlet boundary conditions**

$$h_n(0) = 0 \quad h_n(L) = 0$$

7.0.2 Heat equation on a bounded interval

The goal is to show how tu use Fourier series in order to solve PDEs such as heat equation. Fourier series can be use to solve those equation when **bounded**. When this is not the case, **Fourier transform** will be used in order to solve those equations.

Let us first return to the $1-$dimensional heat equation. This is define as

$$\begin{cases} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f & \forall t > 0, x \in (0, L) \\ u\Big|_{t=0} = u_0 & \text{initial condition} \\ u\Big|_{x=0} = u\Big|_{x=L} = 0 & \text{boundary condition} \end{cases}$$

Let us therefore have an overview over the method to solve those problems

- Decompose the heat equation into Fourier modes
- The PDE becomes an ODE in time for the Fourier coefficients.
- Find a solution by summing the mode solutions

$$\text{PDE} \overset{\text{Fourier}}{\widehat{\longrightarrow}} \text{ODE} \overset{\text{Laplace}}{\widehat{\longrightarrow}} \text{solution}$$

Uniqueness

Theorem 23. The initial value problem for the heat equation has a unique solution.

Proof. *If u_1 and u_2 are two solutions, then $v := u_1 - u_2$ solves*

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ v\mid_{t=0} &= 0 \\ v\mid_{x=0} &= 0 \end{aligned}$$

Again we look for monotone quantity to infer $v = 0$. This quantity here is the following energy

$$\int_0^L v^2(x,t)dx$$

Which is some kind of kinetic energy. Therefore let us try to differentiate this integral w.r.t. t

$$\begin{aligned} \frac{d}{dt}\int_0^L v^2(x,t)dx &= 2\int_0^L v(x,t)\frac{\partial v}{\partial t}(x,t)dx \\ &= 2\int_0^L v(x,t)\frac{\partial^2 v}{\partial x^2}(x,t)dx \end{aligned}$$

Using integration by part

$$= 2\left[v(x,t)\frac{\partial v}{\partial x}(x,t)\right]_0^L - 2\int_0^L \frac{\partial v}{\partial x}(x,t)^2 dx$$
$$= -2\int_0^L \frac{\partial v}{\partial x}(x,t)^2 dx \le 0$$

Using the Dirichlet boundary conditions. Therefore we can infer that the kinetic energy is nonincreasing. Thus

$$0 \le \int_0^L v^2(x,t)dx \le \int_0^L v^2(x,0)dx = 0$$

Therefore, $v(x,t) = 0$ *for every* $t > 0$*, every* $x \in (0, L)$*.*

□

Existence

Let us now take uniqueness for granted, therefore whatever point we use in order to find a solution, if we find a solution then this is the unique solution. On a bounded interval, we'll find it by decomposing the PDE into Fourier modes. The trick we will use is that we'll assume having a solution and then decompose it into Fourier modes in a way that those decomposition respect the boundary conditions.

Since we want $u \mid_{x=0} = u \mid_{x=L} = 0$, we will extend $u(t,x)$ in x into a $2L$-periodic function on $\mathbb{R}$ which is **odd**. In order to make our function odd we need to construct it by taking the symmetry on the left side on the imaginary axis. Then we will do the same for u_0 and f.

> *Remark* Note that if u is a $2L-$periodic and odd function which is continuous, then it necessarily respect Dirichlet boundary conditions.

Ad hoc, for the solution to the heat equation it can be shown: if f is piecewise continuous and u_0 is as well, then for $t > 0, u(t,x)$ and $\frac{\partial u}{\partial x}(f,x)$ are continuous.

Fourier series (respecting $2L-$periodicity and oddness)

Knowing oddness, we know that our function is defined as

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t)\sin\left(\frac{2\pi n}{2L}x\right)$$

b_n needs to be a function of t as it is not fixed for it, meaning that it is defined as

$$b_n(t) = \frac{2}{L}\int_0^L u(x,t)\sin\left(\frac{\pi n}{L}x\right)dx$$

We will do the same for u_0 and f which gives us

$$u_0(x) = \sum_{n=1}^{\infty} b_{0n}\sin\left(\frac{\pi n}{L}x\right)$$
$$f(x,t) = \sum_{n=1}^{\infty} f_n(t)\sin\left(\frac{\pi n}{L}x\right)$$

Now let us differentiate them formally our function u w.r.t. x:

$$\frac{\partial u}{\partial x}(t,x) = \sum_{n=1}^{\infty} b_n(t)\frac{\pi n}{L}\cos\left(\frac{\pi n}{L}x\right)$$
$$\frac{\partial^2 u}{\partial x^2}(t,x) = \sum_{n=1}^{\infty} b_n(t)\left[-\frac{\pi^2 n^2}{L^2}\sin\left(\frac{\pi n}{L}x\right)\right]$$

So now we have one part of the equation done, let us do the other one:

$$\frac{\partial u}{\partial t}(t,x) = \sum_{n=1}^{\infty} b_n'(t)\sin\left(\frac{\pi n}{L}x\right)$$

Since we are looking to find $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f$, this translates into the following

$$\sum_{n=1}^{\infty} \left[b_n'(t) + \frac{\pi^2 n^2}{L^2} b_n(t) \right] \sin\left(\frac{\pi n}{L} x\right) = \sum_{n=1}^{\infty} f_n(t) \sin\left(\frac{\pi n}{L} x\right)$$

By comparison of coefficients we know that for every element n we get the differential equation

$$b_n' - \frac{\pi^2 n^2}{L^2} b_n = f_n \qquad \forall n \geq 1$$

We need to solve infinitely many equations in order to solve this problem. And for the initial condition $u(x,0) = u_0(x)$ which translates into

$$b_n(0) = b_{0n} \qquad \forall n \geq 1$$

Therefore by taking the Fourier mode, we have translate our PDE into infinitely many ODEs that we know how to solve. We can solve it using Laplace transform, this give us

$$b_n(t) = b_{0n} e^{\frac{-\pi^2 n^2}{L^2} t} + \int_0^t e^{-\frac{\pi^2 n^2}{L^2}(t-s)} f_n(s) ds$$

In particular we have found $b_n(t)$ which we can therefore reinserting this into the Fourier expansion of u. This yields the desired solution to the heat equation.

Remark 27.

- if $f = 0$, then $u(x,t) = \sum_{n=1}^{\infty} b_{0n} e^{-\frac{\pi^2 n^2}{L^2} t} \sin\left(\frac{\pi n}{L} x\right)$
- if $u_0 = 0$ and $f(x,t) = \sin\left(\frac{\pi}{L} x\right)$ then we don't need to solve so many ODEs, the solution of these ODEs boils down to a case distinction, meaning that our f_n is defined by the two cases

$$f_n(t) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

  And so

$$b_n(t) = \begin{cases} \frac{L^2}{\pi^2} \left[1 - e^{-\frac{\pi^2}{L^2} t} \right] & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

  And in particular $u(x,t) = \frac{L^2}{\pi^2} \left[1 - e^{-\frac{\pi^2}{L^2} t} \right] \sin\left(\frac{\pi}{L} x\right)$

- If we wanted **Neumann boundary conditions** (namely having $\frac{\partial u}{\partial x}|_{x=0} = 0$ and $\frac{\partial u}{\partial x}|_{x=L} = 0$) we would extend u as an even $2L-$periodic function. This can be kind of intuitive as we would like the derivative to be '*inverse*' (as sin and cos). In this case we would have

$$u(x,t) = \frac{a_0(t)}{2} + \sum_{n=1}^{\infty} a_n(t) \cos\left(\frac{\pi n}{L} x\right)$$

7.0.3 Wave equation on a bounded domain

The definition of wave equation is defined as below

Definition 33. Wave equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = f & \text{for } t > 0, x \in (0, L) \\ u|_{t=0} = u_0 \\ \frac{\partial u}{\partial t}|_{t=0} = u_1 \\ u|_{x=0} = u|_{x=L} = 0 \end{cases}$$

Where condition are initial condition, velocity and boundary condition resp. and c is the speed of propagation.

And as always, we can proof uniqueness of solution.

Uniqueness As always in order to proof uniqueness, the goal is to show that some quantities are constant. In a lot of PDEs those quantities ends up being **energy**.

Proof. *Define*

$$E[u](t) = \frac{1}{2} \int_0^L \underbrace{\left(\frac{\partial u}{\partial t}\right)^2 (x,t)}_{\text{kinetic energy}} + c^2 \underbrace{\left(\frac{\partial u}{\partial x}\right)^2 (x,t)}_{\text{potential energy}} \, dx$$

Then we have

$$\frac{d}{dt} E[u](t) = \int_0^L \frac{\partial u}{\partial t}(x,t) f(x,t) dt$$

Which the proof is left to the reader. This implies the uniqueness of solutions: if u_1 and u_2 are solutions, thus $v := u_1 - u_2$ solves

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} - c^2 \frac{\partial^2 v}{\partial x^2} = 0 \\ v|_{t=0} = 0 \\ \frac{\partial v}{\partial t}|_{t=0} = 0 \\ v|_{x=0} = v|_{x=L} = 0 \end{cases}$$

By conservation of energy, $E[u](t)$ is constant in t (i.e. $E[u](t) = E[u](0)$). But $v|_{t=0}$ and $\frac{\partial v}{\partial t}|_{t=0}$ both vanish, therefore $E[u](0) = 0$. In term, this means that

$$\frac{\partial v}{\partial t} = 0 \text{ and } \frac{\partial v}{\partial x} = 0$$

Hence v is constant and equal to 0. □

Existence **Proof.** *Let us extend $u(x,t), f(x,t), u_0(x), u_1(x)$ as an odd $2L-$periodic function on all of $x \in \mathbb{R}$. Then Fourier series will only contain* sin *terms as follows*

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{\pi n}{L} x\right)$$

$$u_0(x) = \sum_{n=1}^{\infty} b_{0n} \sin\left(\frac{\pi n}{L} x\right)$$

$$u_1(x) = \sum_{n=1}^{\infty} b_{1n} \sin\left(\frac{\pi n}{L} x\right)$$

$$f(x,t) = \sum_{n=1}^{\infty} f_n(t) \sin\left(\frac{\pi n}{L} x\right)$$

Now we again look at the wave equation and compute all relevant derivatives.

$$\frac{\partial^2 u}{\partial t^2}(x,t) = \sum_{n=1}^{\infty} b_n''(t)\sin\Big(\frac{\pi n}{L}x\Big)$$
$$\frac{\partial^2 u}{\partial x^2}(x,t) = \sum_{n=1}^{\infty} b_n(t)\Big(-\frac{\pi^2 n^2}{L^2}\Big)\sin\Big(\frac{\pi n}{L}x\Big)$$

Now again we insert this into the PDEs which gives us

$$\sum_{n=1}^{\infty}\Big[b_n''(t) + c^2\frac{\pi^2 n^2}{L^2}b_n(t)\Big]\sin\Big(\frac{\pi n}{L}x\Big) = \sum_{n=1}^{\infty} f_n(t)\sin\Big(\frac{\pi n}{L}x\Big)$$

By comparing the coefficients we get

$$b_n'' + c^2\frac{\pi^2 n^2}{L^2}b_n = f_n$$
$$b_n(0) = b_{0n}$$
$$b_n'(0) = b_{in}$$

And this has to hold for every $n \geq 1$.

We can solve this ODE by using Laplace transform and get the closed form

$$b_n(t) = b_{0n}\cos\Big(\frac{c\pi n}{L}t\Big) + \frac{Lb_{1n}}{c\pi n}\sin\Big(\frac{c\pi n}{L}t\Big) + \int_0^t \frac{L}{c\pi n}\sin\Big(\frac{c\pi n}{L}(t-s)\Big)f(s)ds$$

□

Remark 28. The above methods apply to more general initial and boundary value problems for linear PDEs when $t > 0, x \in (0, L)$ (**bounded domain**).

1. We '*need*' the coefficients of the PDE to be constant. (it works beyond that, but the resulting ODEs becomes very nasty)
2. One can address more general classes of boundary conditions, e.g. periodic boundary conditions
3. Inhomogenous boundary conditions
4. More generally, initial boundary problems of the form

$$\frac{\partial u}{\partial t} + P_u = 0$$
$$u|_{t=0} = u|_{x=L} = 0$$

This requires decomposing $u(x,t)$ into a sum of eigenfunctions of the operator P (Sturm-Liouville problem).